

PUBLISHED: 16-FEB-2017

2014.0 RANGE ROVER (LG), 100-00

# GENERAL INFORMATION

**POWERTRAIN CONTROL MODULE (PCM) - B10A2-31 TO P0430-00 - 5.0 L PETROL****WARNING:**

Fuel injector voltage will reach 65 Volts during operation and have a high current requirement.

**CAUTION:**

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

**NOTES:**

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.

- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Powertrain Control Module (PCM) - 5.0L Petrol. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: Electronic Engine Controls (303-14 Electronic Engine Controls - V8 S/C 5.0L Petrol, Diagnosis and Testing).

| DTC      | DESCRIPTION                             | POSSIBLE CAUSES                                                                                                                                                                                                                                                                                                   | ACTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B10A2-31 | Crash Input - No signal                 | <ul style="list-style-type: none"> <li>■ The powertrain control module input crash signal circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Restraints control module power or ground connection failure</li> <li>■ Restraints control module failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check the powertrain control module input crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Refer to the electrical circuit diagrams and check the power and ground connections to the module</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related DTCs and refer to the relevant DTC index</li> </ul>                |
| B10A2-32 | Crash Input - Signal low time < minimum | <ul style="list-style-type: none"> <li>■ This DTC is set when the engine control module detects that the signal low time is less than the minimum value</li> <li>■ Restraints control module failure</li> <li>■ Central junction box failure</li> <li>■ Harness failure</li> </ul>                                | <ul style="list-style-type: none"> <li>■ This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the engine control module</li> <li>■ Check the restraints control module for DTCs and refer to the relevant DTC index</li> <li>■ Check central junction box for DTCs and refer to the relevant DTC index</li> <li>■ Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a</li> </ul> |

|          |                                          |                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                          |                                                                                                                                                                                                                                                                                        | <p>single wire which connects the restraints control module to the central junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| B10A2-35 | Crash Input - Signal high time > maximum | <ul style="list-style-type: none"> <li>■ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value</li> <li>■ Restraints control module failure</li> <li>■ Central junction box failure</li> <li>■ Harness failure</li> </ul> | <ul style="list-style-type: none"> <li>■ This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the engine control module</li> <li>■ Check the restraints control module for DTCs and refer to the relevant DTC index</li> <li>■ Check central junction box for DTCs and refer to the relevant DTC index</li> <li>■ Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the central junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest</li> </ul> |
| B10A2-36 | Crash Input - Signal frequency too low   | <ul style="list-style-type: none"> <li>■ Restraints control module failure</li> <li>■ Body control module failure</li> <li>■ Harness failure</li> </ul>                                                                                                                                | <ul style="list-style-type: none"> <li>■ This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the powertrain control module</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related DTCs and refer to the relevant DTC index</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment, check the body</li> </ul>                                                                                                                                                                                                                                                                                                                               |

|          |                                         |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                         |                                                                                                                                                                                  | <p>control module for related DTCs and refer to the relevant DTC index</p> <ul style="list-style-type: none"> <li>Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the body control module and the powertrain control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      |
| B10A2-37 | Crash Input - Signal frequency too high | <ul style="list-style-type: none"> <li>Restraints control module failure</li> <li>Body control module failure</li> <li>Harness failure</li> </ul>                                | <ul style="list-style-type: none"> <li>This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the powertrain control module</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, check the restraints control module for related DTCs and refer to the relevant DTC index</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, check the body control module for related DTCs and refer to the relevant DTC index</li> <li>Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the body control module and the powertrain control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Clear DTC and retest</li> </ul> |
| B1206-68 | Crash Occurred - Event information      | <ul style="list-style-type: none"> <li>This DTC is set if the restraints control module has deployed the restraints systems following activation of the crash sensors</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check the restraints control module for DTCs and refer to the relevant DTC index</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|          |                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                         | <ul style="list-style-type: none"> <li>Event information. The engine control module has received a crash signal from the restraints control module</li> </ul>                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| B1A75-00 | Fuel Sender No.1 - No sub type information              | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Within the engine control module software a fuel level model is created. If the error between the fuel level model and the measured fuel level has gone over a threshold, and no change in fuel level is detected a DTC is set</p> </div> <ul style="list-style-type: none"> <li>The fuel tank active side fuel level sensor is stuck</li> </ul>  | <ul style="list-style-type: none"> <li>Check the fuel level sensor for correct movement across the entire range</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                |
| B1A76-00 | Fuel Sender No.2 - No sub type information              | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Within the engine control module software a fuel level model is created. If the error between the fuel level model and the measured fuel level has gone over a threshold, and no change in fuel level is detected a DTC is set</p> </div> <ul style="list-style-type: none"> <li>The fuel tank passive side fuel level sensor is stuck</li> </ul> | <ul style="list-style-type: none"> <li>Check the fuel level sensor for correct movement across the entire range</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                |
| C0031-00 | Left Front Wheel Speed Sensor - No sub type information | <ul style="list-style-type: none"> <li>Anti-lock brake system control module, has indicated a failure with wheel speed sensor signal</li> <li>Wheel speed sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Wheel speed sensor failure</li> <li>Reluctor ring damaged</li> </ul>                                                                                                                                                             | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index</li> <li>Refer to the electrical circuit diagrams and check the wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install a new wheel speed sensor as required</li> <li>Inspect reluctor ring for damage</li> </ul> |

|          |                                                          |                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                          |                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| C0034-00 | Right Front Wheel Speed Sensor - No sub type information | <ul style="list-style-type: none"> <li>Anti-lock brake system control module, has indicated a failure with wheel speed sensor signal</li> <li>Wheel speed sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Wheel speed sensor failure</li> <li>Reluctor ring damaged</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index</li> <li>Refer to the electrical circuit diagrams and check the wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install a new wheel speed sensor as required</li> <li>Inspect reluctor ring for damage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> </ul> |
| C0037-00 | Left Rear Wheel Speed Sensor - No sub type information   | <ul style="list-style-type: none"> <li>Anti-lock brake system control module, has indicated a failure with wheel speed sensor signal</li> <li>Wheel speed sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Wheel speed sensor failure</li> <li>Reluctor ring damaged</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index</li> <li>Refer to the electrical circuit diagrams and check the wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install a new wheel speed sensor as required</li> <li>Inspect reluctor ring for damage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> </ul> |
| C003A-00 | Right Rear Wheel Speed                                   | <ul style="list-style-type: none"> <li>Anti-lock brake system control module, has indicated a failure with wheel speed sensor signal</li> </ul>                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|          |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Sensor - No sub type information       | <ul style="list-style-type: none"> <li>Wheel speed sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Wheel speed sensor failure</li> <li>Reluctor ring damaged</li> </ul>                                                                                                                                                                                              | <p>module for related DTCs and refer to the relevant DTC index</p> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the wheel speed sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install a new wheel speed sensor as required</li> <li>Inspect reluctor ring for damage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| C1001-87 | Vision System Camera - Missing message | <ul style="list-style-type: none"> <li>Image processing module power or ground circuit open circuit, high resistance</li> <li>High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>FlexRay bus circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Intelligent emergency braking system fault</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the image processing module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary</li> <li>Refer to the electrical circuit diagrams and check the FlexRay bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, check the image processing module for related DTCs and refer to the relevant DTC index</li> </ul> |
| P000A-00 | Intake (A) Camshaft Position Slow      | <div>  <b>NOTE:</b> </div>                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Check the oil level is correct</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|          |                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Response Bank 1 - No sub type information                                    | <p><b>Monitor description.</b> Inlet camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction</li> </ul>                                                                                                                                         | <ul style="list-style-type: none"> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul> |
| P000B-00 | Exhaust (B) Camshaft Position Slow Response Bank 1 - No sub type information | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Exhaust camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                      |

|          |                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P000C-00 | Intake (A) Camshaft Position Slow Response Bank 2 - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Inlet camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> </div> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul> |
| P000D-00 | Exhaust (B) Camshaft Position Slow Response Bank 2 - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Exhaust camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> </div> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> </ul>                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|          |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                         | <ul style="list-style-type: none"> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul>                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul>                                                                                  |
| P0010-00 | Intake (A) Camshaft Position Actuator Circuit / Open (Bank 1) - No sub type information | <ul style="list-style-type: none"> <li>Variable camshaft timing solenoid circuit open circuit, high resistance</li> <li>Variable camshaft timing solenoid failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing solenoid as required</li> </ul>                                                                                                                                                                                                                                              |
| P0011-00 | Intake (A) Camshaft Position Timing - Over-Advanced (Bank 1) - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Inlet camshaft is unable to achieve the desired set point angle. The camshaft is stuck         </div> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic</li> </ul> |

|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>▪ Check and install new variable camshaft timing actuator as required</li> <li>▪ Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| P0011-71 | Intake (A) Camshaft Position Timing - Over-Advanced (Bank 1) - Actuator stuck | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Inlet camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> </div> <ul style="list-style-type: none"> <li>▪ Oil level is low</li> <li>▪ Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>▪ Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Oil pressure is low</li> <li>▪ Variable camshaft timing actuator mechanical failure</li> <li>▪ Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>▪ Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>▪ Oil temperature greater than -10 °C, ensure no other cam or oil temperature related faults are present, drive vehicle in part load conditions such that the inlet variable valve timing would normally be active, hold various speed load conditions for 6 to 10 seconds, conduct at least 5 load changes</li> </ul> </li> <li>▪ Prioritised Checks to Perform</li> <li>▪ Check the oil level is correct</li> <li>▪ Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>▪ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P0013-00 | Exhaust (B) Camshaft Position Actuator Circuit / Open (Bank 1) - No sub type information | <ul style="list-style-type: none"> <li>Variable camshaft timing solenoid circuit open circuit, high resistance</li> <li>Variable camshaft timing solenoid failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing solenoid as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P0014-00 | Exhaust (B) Camshaft Position Timing - Over-Advanced (Bank 1) - No sub type information  | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> Exhaust camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> </div> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul> |

|          |                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0014-71 | Exhaust (B) Camshaft Position Timing - Over-Advanced (Bank 1) Actuator stuck | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Exhaust camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> </div> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>Oil temperature greater than -10 °C, ensure no other cam or oil temperature related faults are present, drive vehicle in part load conditions such that the inlet variable valve timing would normally be active, hold various speed load conditions for 6 to 10 seconds, conduct at least 5 load changes</li> </ul> </li> <li>Prioritised Checks to Perform</li> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul> |
| P0016-00 | Crankshaft Position - Camshaft Position Correlation - Bank 1                 | <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Loose crankshaft position sensor</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Check crankshaft position sensor is installed correctly</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Sensor A - No sub type information                                                                 | <ul style="list-style-type: none"> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft or crankshaft position sensor reluctor ring air gap excessive</li> <li>Timing chain stretch beyond a tolerable limit</li> <li>Valve timing incorrect</li> </ul>                                                                                               | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Check crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> <li>Refer to the relevant section of the workshop manual and check valve timing is within specification</li> </ul> |
| P0016-76 | Crankshaft Position - Camshaft Position Correlation - Bank 1 Sensor A - Wrong mounting position    | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Initial reference adaption of cam wheel edges to determine system is OK post engine rebuild or valve timing overhaul</p> </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Loose crankshaft position sensor</li> <li>Camshaft or crankshaft position sensor reluctor ring air gap excessive</li> <li>Engine assembled incorrectly</li> <li>Valve timing incorrect</li> </ul> | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Check crankshaft position sensor is installed correctly</li> <li>Refer to the relevant section of the workshop manual and check valve timing is within specification</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul>                                                                                                                                                                              |
| P0016-79 | Crankshaft Position - Camshaft Position Correlation - Bank 1 Sensor A - Mechanical linkage failure | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Correct engagement of camshaft locking pin. The average camshaft position is outside of the allowable limit during engine start. The engine will start poorly</p> </div> <ul style="list-style-type: none"> <li>Camshaft locking pin is not engaged during start</li> <li>Loose camshaft position sensor</li> </ul>                                                                                            | <ul style="list-style-type: none"> <li>Check variable camshaft timing actuator for mechanical damage</li> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to variable camshaft timing actuator is correct and that oil gallery is free from debris</li> <li>Refer to the relevant sections of the workshop manual and check the</li> </ul>                                                               |

|          |                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                 | <ul style="list-style-type: none"> <li>Camshaft position sensor disturbed signal</li> <li>Oil pressure to variable camshaft timing actuator is low</li> <li>Camshaft chain tensioner failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                             | camshaft chain tensioner is operating correctly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| P0017-00 | Crankshaft Position - Camshaft Position Correlation - Bank 1 Sensor B - No sub type information | <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Loose crankshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft or crankshaft position sensor reluctor ring air gap excessive</li> <li>Timing chain stretch beyond a tolerable limit</li> <li>Valve timing incorrect</li> </ul>               | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Check crankshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Check crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> <li>Refer to the relevant section of the workshop manual and check valve timing is within specification</li> </ul> |
| P0017-76 | Crankshaft Position - Camshaft Position Correlation - Bank 1 Sensor B - Wrong mounting position | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Initial reference adaption of cam wheel edges to determine system is OK post engine rebuild or valve timing overhaul</p> </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Loose crankshaft position sensor</li> <li>Camshaft or crankshaft position sensor reluctor ring air gap excessive</li> <li>Engine assembled incorrectly</li> <li>Valve timing incorrect</li> </ul> | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Check crankshaft position sensor is installed correctly</li> <li>Refer to the relevant section of the workshop manual and check valve timing is within specification</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul>                                                                                                                                                                                                                                                                                                              |
| P0017-79 | Crankshaft Position - Camshaft Position Correlation -                                           | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Correct engagement of camshaft</p> </div>                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Check variable camshaft timing actuator for mechanical damage</li> <li>Check camshaft position sensor is installed correctly</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Bank 1<br>Sensor B -<br>Mechanical<br>linkage failure                                              | <p>locking pin. The average camshaft position is outside of the allowable limit during engine start. The engine will start poorly</p> <ul style="list-style-type: none"> <li>Camshaft locking pin is not engaged during start</li> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor disturbed signal</li> <li>Oil pressure to variable camshaft timing actuator is low</li> <li>Camshaft chain tensioner failure</li> </ul>                                                                                                       | <ul style="list-style-type: none"> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to variable camshaft timing actuator is correct and that oil gallery is free from debris</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft chain tensioner is operating correctly</li> </ul>                                                                                                                                                                                                                                                                                     |
| P0018-00 | Crankshaft Position - Camshaft Position Correlation - Bank 2<br>Sensor A - No sub type information | <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Loose crankshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft or crankshaft position sensor reluctor ring air gap excessive</li> <li>Timing chain stretch beyond a tolerable limit</li> <li>Valve timing incorrect</li> </ul> | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Check crankshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Check crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> <li>Refer to the relevant section of the workshop manual and check valve timing is within specification</li> </ul> |
| P0018-76 | Crankshaft Position - Camshaft Position Correlation - Bank 2<br>Sensor A - wrong mounting position | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Initial reference adaption of cam wheel edges to determine system is OK post engine rebuild or valve timing overhaul</p> </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Loose crankshaft position sensor</li> </ul>                                                                                                                                         | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Check crankshaft position sensor is installed correctly</li> <li>Refer to the relevant section of the workshop manual and check valve timing is within specification</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul>                                                                                                                                                                                                                                                                                                              |

|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                    | <ul style="list-style-type: none"> <li>Camshaft or crankshaft position sensor reluctor ring air gap excessive</li> <li>Engine assembled incorrectly</li> <li>Valve timing incorrect</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| P0018-79 | Crankshaft Position - Camshaft Position Correlation - Bank 2 Sensor A - mechanical linkage failure | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Correct engagement of camshaft locking pin. The average camshaft position is outside of the allowable limit during engine start. The engine will start poorly</p> </div> <ul style="list-style-type: none"> <li>Camshaft locking pin is not engaged during start</li> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor disturbed signal</li> <li>Oil pressure to variable camshaft timing actuator is low</li> <li>Camshaft chain tensioner failure</li> </ul> | <ul style="list-style-type: none"> <li>Check variable camshaft timing actuator for mechanical damage</li> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to variable camshaft timing actuator is correct and that oil gallery is free from debris</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft chain tensioner is operating correctly</li> </ul>                                                                                                                                               |
| P0019-00 | Crankshaft Position - Camshaft Position Correlation - Bank 2 Sensor B - No sub type information    | <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Loose crankshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft or crankshaft position sensor reluctor ring air gap excessive</li> <li>Timing chain stretch beyond a tolerable limit</li> <li>Valve timing incorrect</li> </ul>                                                                               | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Check crankshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Check crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> <li>Refer to the relevant section of the workshop manual and check valve timing is within specification</li> </ul> |
| P0019-76 | Crankshaft Position - Camshaft                                                                     | <div>  <b>NOTE:</b> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Position Correlation - Bank 2 Sensor B - Wrong mounting position                                   | <p><b>Monitor description.</b> Initial reference adaption of cam wheel edges to determine system is OK post engine rebuild or valve timing overhaul</p> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Loose crankshaft position sensor</li> <li>Camshaft or crankshaft position sensor reluctor ring air gap excessive</li> <li>Engine assembled incorrectly</li> <li>Valve timing incorrect</li> </ul>                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Check crankshaft position sensor is installed correctly</li> <li>Refer to the relevant section of the workshop manual and check valve timing is within specification</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul>                                                                                                                                                                                                                               |
| P0019-79 | Crankshaft Position - Camshaft Position Correlation - Bank 2 Sensor B - Mechanical linkage failure | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Correct engagement of camshaft locking pin. The average camshaft position is outside of the allowable limit during engine start. The engine will start poorly</p> <ul style="list-style-type: none"> <li>Camshaft locking pin is not engaged during start</li> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor disturbed signal</li> <li>Oil pressure to variable camshaft timing actuator is low</li> <li>Camshaft chain tensioner failure</li> </ul> | <ul style="list-style-type: none"> <li>Check variable camshaft timing actuator for mechanical damage</li> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to variable camshaft timing actuator is correct and that oil gallery is free from debris</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft chain tensioner is operating correctly</li> </ul> |
| P001A-00 | Intake (A) Cam Profile Control Circuit / Open (Bank 1) - No sub type information                   | <ul style="list-style-type: none"> <li>Camshaft profile switching solenoid circuit, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                       |
| P001B-00 | Intake (A) Cam Profile                                                                             | <ul style="list-style-type: none"> <li>Camshaft profile switching solenoid circuit short circuit to ground</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|          |                                                                                   |                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Control Circuit Low (Bank 1)<br>- No sub type information                         |                                                                                                                              | <p>camshaft profile switching solenoid circuit for short circuit to ground</p> <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                            |
| P001C-00 | Intake (A) Cam Profile Control Circuit High (Bank 1)<br>- No sub type information | <ul style="list-style-type: none"> <li>Camshaft profile switching solenoid circuit short circuit to power</li> </ul>         | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for short circuit to power</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>        |
| P001D-00 | Intake (A) Cam Profile Control Circuit / Open (Bank 2) - No sub type information  | <ul style="list-style-type: none"> <li>Camshaft profile switching solenoid circuit, open circuit, high resistance</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P001E-00 | Intake (A) Cam Profile Control Circuit Low (Bank 2)<br>- No sub type information  | <ul style="list-style-type: none"> <li>Camshaft profile switching solenoid circuit short circuit to ground</li> </ul>        | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for short circuit to ground</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>       |

|          |                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P001F-00 | Intake (A)<br>Cam Profile<br>Control Circuit<br>High (Bank 2)<br>- No sub type<br>information                | <ul style="list-style-type: none"> <li>Camshaft profile switching solenoid circuit short circuit to power</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for short circuit to power</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                             |
| P0020-00 | Intake (A)<br>Camshaft<br>Position<br>Actuator<br>Circuit / Open<br>(Bank 2) - No<br>sub type<br>information | <ul style="list-style-type: none"> <li>Variable camshaft timing solenoid circuit open circuit, high resistance</li> <li>Variable camshaft timing solenoid failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing solenoid as required</li> </ul>                                                                                                                                                                                                                                                                                                                        |
| P0021-00 | Intake (A)<br>Camshaft<br>Position<br>Timing - Over-<br>Advanced<br>(Bank 2) - No<br>sub type<br>information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Inlet camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> </div> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |

|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>▪ Check and install new variable camshaft timing actuator as required</li> <li>▪ Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| P0021-71 | Intake (A) Camshaft Position Timing - Over-Advanced (Bank 2) - Actuator stuck | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Inlet camshaft is unable to achieve the desired set point angle. The camshaft is stuck</p> </div> <ul style="list-style-type: none"> <li>▪ Oil level is low</li> <li>▪ Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>▪ Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Oil pressure is low</li> <li>▪ Variable camshaft timing actuator mechanical failure</li> <li>▪ Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>▪ Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>▪ Oil temperature greater than -10 °C, ensure no other cam or oil temperature related faults are present, drive vehicle in part load conditions such that the inlet variable valve timing would normally be active, hold various speed load conditions for 6 to 10 seconds, conduct at least 5 load changes</li> </ul> </li> <li>▪ Prioritised Checks to Perform</li> <li>▪ Check the oil level is correct</li> <li>▪ Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>▪ Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new variable camshaft timing actuator as required</li> <li>▪ Refer to the relevant sections of the workshop</li> </ul> |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | manual and check the camshaft operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| P0023-00 | Exhaust (B) Camshaft Position Actuator Circuit / Open (Bank 2) - No sub type information | <ul style="list-style-type: none"> <li>Variable camshaft timing solenoid circuit open circuit, high resistance</li> <li>Variable camshaft timing solenoid failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing solenoid as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P0024-00 | Exhaust (B) Camshaft Position Timing - Over-Advanced (Bank 2) - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Exhaust camshaft is unable to achieve the desired set point angle. The camshaft is stuck         </div> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul> | <ul style="list-style-type: none"> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul> |
| P0024-71 | Exhaust (B) Camshaft Position Timing - Over-                                             | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Exhaust camshaft is unable to achieve         </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy             <ul style="list-style-type: none"> <li>Oil temperature greater than -10 °C, ensure no other cam</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|          |                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Advanced<br>(Bank 2) -<br>Actuator stuck                                    | <div>the desired set point angle.<br/>The camshaft is stuck</div> <ul style="list-style-type: none"> <li>Oil level is low</li> <li>Variable camshaft timing solenoid circuit intake and exhaust connectors swapped</li> <li>Variable camshaft timing solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil pressure is low</li> <li>Variable camshaft timing actuator mechanical failure</li> <li>Excessive camshaft friction, restricted movement or stuck</li> </ul> | <p>or oil temperature related faults are present, drive vehicle in part load conditions such that the inlet variable valve timing would normally be active, hold various speed load conditions for 6 to 10 seconds, conduct at least 5 load changes</p> <ul style="list-style-type: none"> <li>Prioritised Checks to Perform</li> <li>Check the oil level is correct</li> <li>Check variable camshaft timing solenoid circuit intake and exhaust connectors are not swapped</li> <li>Refer to the electrical circuit diagrams and check variable camshaft timing solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new variable camshaft timing actuator as required</li> <li>Refer to the relevant sections of the workshop manual and check the camshaft operation</li> </ul> |
| P0030-00 | HO2S Heater Control Circuit (Bank 1, Sensor 1) -<br>No sub type information | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Heated oxygen sensor heater duty cycle maintains a constant value         </div>                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                              | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                                                                                                                              |
| P0031-00 | HO2S Heater Control Circuit Low (Bank 1, Sensor 1) - No sub type information                 | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Heated oxygen sensor heater duty cycle below a threshold         </div> <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul>   | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0032-00 | HO2S Heater Control Circuit High (Bank 1, Sensor 1) - No sub type information                | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Heated oxygen sensor heater duty cycle above a threshold         </div> <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0033-00 | Turbocharger /Supercharger Bypass Valve "A" Control Circuit / Open - No sub type information | <ul style="list-style-type: none"> <li>▪ Supercharger bypass valve actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Supercharger bypass valve actuator failure</li> </ul>                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check supercharger bypass valve actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                             |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                          |                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new supercharger bypass valve actuator as required</li> </ul>                                                                                                                                                                                           |
| P0034-00 | Turbocharger /Supercharger Bypass Valve "A"Control Circuit Low - No sub type information | <ul style="list-style-type: none"> <li>Supercharger bypass valve actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Supercharger bypass valve actuator failure</li> </ul>                                                                     | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check supercharger bypass valve actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new supercharger bypass valve actuator as required</li> </ul> |
| P0036-00 | HO2S Heater Control Circuit (Bank 1, Sensor 2) - No sub type information                 | <ul style="list-style-type: none"> <li>Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                               |
| P0037-00 | HO2S Heater Control Circuit Low (Bank 1, Sensor 2) - No sub type information             | <ul style="list-style-type: none"> <li>Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                       | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                      |

|          |                                                                                                          |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                          | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                                                                                                                                                                                                      |
| P0038-00 | HO2S Heater Control Circuit High (Bank 1, Sensor 2) - No sub type information                            | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                 |
| P0039-00 | Turbocharger /Supercharger Bypass Valve "A" Control Circuit Range /Performance - No sub type information | <ul style="list-style-type: none"> <li>▪ Supercharger bypass valve actuator circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Supercharger bypass valve actuator failure</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check supercharger bypass valve actuator circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new supercharger bypass valve actuator as required</li> </ul> |
| P003A-21 | Turbocharger /Supercharger Boost Control A Position Exceeded Learning Limit - Signal amplitude < minimum | <ul style="list-style-type: none"> <li>▪ Supercharger mechanical end stop incorrect adjustment</li> <li>▪ Supercharger bypass blade and end stop assembly failure</li> <li>▪ Supercharger actuator failure</li> </ul>                                                                                     | <ul style="list-style-type: none"> <li>▪ Inspect supercharger mechanical end stop for correct adjustment</li> <li>▪ Check for correct assembly of bypass blade and end stop</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored</li> </ul>                                                                                                                                                                                                                                                                             |

|          |                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>Check and install new supercharger actuator as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| P003A-22 | Turbocharger /Supercharger Boost Control A Position Exceeded Learning Limit - Signal amplitude > maximum | <ul style="list-style-type: none"> <li>Supercharger mechanical end stop incorrect adjustment</li> <li>Bypass blade restriction due to debris or fouling</li> <li>Supercharger bypass blade and end stop assembly failure</li> <li>Supercharger actuator failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Inspect supercharger mechanical end stop for correct adjustment</li> <li>Check for correct assembly of bypass blade and for sign of restriction due to debris or fouling</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new supercharger actuator as required</li> </ul>                                                                                                                                                                                                                                                                                                                                          |
| P003C-00 | A Camshaft Profile Control Performance /Stuck Off (Bank 1) - No sub type information                     | <ul style="list-style-type: none"> <li>Cylinder head oil pressure too high /low</li> <li>Camshaft profile switching solenoid oil gallery feed blocked</li> <li>Camshaft profile switching solenoid oil gallery drain blocked</li> <li>Camshaft profile switching solenoid oil gallery restrictor missing, damaged or incorrectly installed</li> <li>Multiple camshaft profile switching solenoid tappet failures</li> <li>Heated oxygen sensor slow response</li> <li>Other fuel system failures</li> <li>Other variable camshaft timing failures</li> <li>Camshaft profile switching solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft profile switching solenoid failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install new camshaft profile switching solenoid as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P003C-77 | A Camshaft Profile Control Performance /Stuck Off (Bank 1) - Commanded position not reachable            | <ul style="list-style-type: none"> <li>Cylinder head oil pressure too high /low</li> <li>Camshaft profile switching solenoid oil gallery feed blocked</li> <li>Camshaft profile switching solenoid oil gallery drain blocked</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|          |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                               | <ul style="list-style-type: none"> <li>Camshaft profile switching solenoid oil gallery restrictor missing, damaged or incorrectly installed</li> <li>Multiple camshaft profile switching solenoid tappet failures</li> <li>Heated oxygen sensor slow response</li> <li>Other fuel system failures</li> <li>Other variable camshaft timing failures</li> <li>Camshaft profile switching solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft profile switching solenoid failure</li> </ul>                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install new camshaft profile switching solenoid as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                     |
| P003E-00 | A Camshaft Profile Control Performance/ Stuck Off (Bank 2) - No sub type information          | <ul style="list-style-type: none"> <li>Cylinder head oil pressure too high /low</li> <li>Camshaft profile switching solenoid oil gallery feed blocked</li> <li>Camshaft profile switching solenoid oil gallery drain blocked</li> <li>Camshaft profile switching solenoid oil gallery restrictor missing, damaged or incorrectly installed</li> <li>Multiple camshaft profile switching solenoid tappet failures</li> <li>Heated oxygen sensor slow response</li> <li>Other fuel system failures</li> <li>Other variable camshaft timing failures</li> <li>Camshaft profile switching solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft profile switching solenoid failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install new camshaft profile switching solenoid as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P003E-77 | A Camshaft Profile Control Performance/ Stuck Off (Bank 2) - Commanded position not reachable | <ul style="list-style-type: none"> <li>Cylinder head oil pressure too high /low</li> <li>Camshaft profile switching solenoid oil gallery feed blocked</li> <li>Camshaft profile switching solenoid oil gallery drain blocked</li> <li>Camshaft profile switching solenoid oil gallery restrictor missing, damaged or incorrectly installed</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Refer to the relevant sections of the workshop manual and check the oil pressure is within specification. Check oil supply to camshaft is correct and that oil gallery is free from debris</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|          |                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                             | <ul style="list-style-type: none"> <li>▪ Multiple camshaft profile switching solenoid tappet failures</li> <li>▪ Heated oxygen sensor slow response</li> <li>▪ Other fuel system failures</li> <li>▪ Other variable camshaft timing failures</li> <li>▪ Camshaft profile switching solenoid circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Camshaft profile switching solenoid failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>▪ Refer to the electrical circuit diagrams and check camshaft profile switching solenoid circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Check and install new camshaft profile switching solenoid as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0040-00 | Oxygen Sensor Signals Swapped Bank 1 / Sensor 1 / Bank 2 Sensor 1 - No sub type information | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor is installed in the wrong bank</li> </ul>                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>▪ Check and install the heated oxygen sensor to the correct position in the exhaust system</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P0041-00 | Oxygen Sensor Signals Swapped Bank 1 / Sensor 2 / Bank 2 Sensor 2 - No sub type information | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor is installed in the wrong bank</li> </ul>                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>▪ Check and install the heated oxygen sensor to the correct position in the exhaust system</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P0042-00 | HO2S Heater Control Circuit (Bank 1, Sensor 3) - No sub type information                    | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul>                                                                                                                                            | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                                    |
| P0043-   | HO2S Heater                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|          |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00       | Control Circuit Low (Bank 1, Sensor 3) - No sub type information                                  | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul>                                         | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0044-00 | HO2S Heater Control Circuit High (Bank 1, Sensor 3) - No sub type information                     | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul>                                         | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P004B-00 | Turbocharger /Supercharger Boost Control "B" Circuit Range /Performance - No sub type information | <ul style="list-style-type: none"> <li>■ Charge air coolant pump circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Charge air coolant pump relay failure</li> </ul>                                                                                                                              | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check the charge air coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Check and install a new charge air coolant pump relay as required</li> </ul>                                                                                                                                                               |
| P0050-00 | HO2S Heater Control Circuit (Bank 2, Sensor 1) - No sub type information                          | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Heated oxygen sensor heater duty cycle maintains a constant value         </div> <ul style="list-style-type: none"> <li>■ Heated oxygen sensor heater control circuit short circuit to</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                     |

|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                               | <p>ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                                                                                                                                                                                                                         |
| P0051-00 | HO2S Heater Control Circuit Low (Bank 2, Sensor 1) - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Heated oxygen sensor heater duty cycle below a threshold</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>   | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0052-00 | HO2S Heater Control Circuit High (Bank 2, Sensor 1) - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Heated oxygen sensor heater duty cycle above a threshold</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor heater control circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0054-00 | HO2S Heater Resistance (Bank 1, Sensor 2) - No sub type information           | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Heated oxygen sensor heater not heating enough</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                  | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L -</li> </ul>                                                                                        |

|          |                                                                              |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                              | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul>                                                                                                                               | Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A                                                                                                                                                                                                                                                                                                                                                                                           |
| P0055-00 | HO2S Heater Resistance (Bank 1, Sensor 3) - No sub type information          | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0056-00 | HO2S Heater Control Circuit (Bank 2, Sensor 2) - No sub type information     | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0057-00 | HO2S Heater Control Circuit Low (Bank 2, Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |

|          |                                                                               |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0058-00 | HO2S Heater Control Circuit High (Bank 2, Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0060-00 | HO2S Heater Resistance (Bank 2, Sensor 2) - No sub type information           | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0061-00 | HO2S Heater Resistance (Bank 2, Sensor 3) - No sub type information           | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0062-00 | HO2S Heater Control Circuit (Bank 2, Sensor 3) -                              | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short</li> </ul>                                                                                                                                                                                                                                                                                                                     |

|          |                                                                               |                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | No sub type information                                                       | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                             | <p>circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                                    |
| P0063-00 | HO2S Heater Control Circuit Low (Bank 2, Sensor 3) - No sub type information  | <ul style="list-style-type: none"> <li>Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0064-00 | HO2S Heater Control Circuit High (Bank 2, Sensor 3) - No sub type information | <ul style="list-style-type: none"> <li>Heated oxygen sensor heater circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0069-64 | MAP - Barometric Pressure Correlation - Signal plausibility failure           | <div>  <b>NOTE:</b> </div> <div> <p>The sensor is installed within the engine control module and is not serviceable</p> </div>                                                                                   | <ul style="list-style-type: none"> <li>Check and install new engine control module as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> </ul>                                                                                                                                                                                                                 |

|          |                                                                                            |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                            | <ul style="list-style-type: none"> <li>Engine control module failure</li> </ul>                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| P0070-00 | Ambient Air Temperature Sensor Circuit "A" - No sub type information                       | <ul style="list-style-type: none"> <li>Ambient air temperature sensor, (usually located in the door mirror) circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Ambient air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) <ul style="list-style-type: none"> <li>Check that a plausible value is displayed</li> </ul> </li> <li>Refer to the electrical circuit diagrams and check ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new ambient air temperature sensor as required</li> </ul> |
| P0071-21 | Ambient Air Temperature Sensor Circuit "A" Range /Performance - Signal amplitude < minimum | <ul style="list-style-type: none"> <li>Rapid temperature change of ambient air temperature sensor</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Ambient air temperature sensor failure</li> </ul>                                                                                                  | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) <ul style="list-style-type: none"> <li>Check that a plausible value is displayed</li> </ul> </li> <li>Refer to the electrical circuit diagrams and check ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new ambient air temperature sensor as required</li> </ul> |
|          |                                                                                            |                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|          |                                                                                            |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0071-22 | Ambient Air Temperature Sensor Circuit "A" Range /Performance - Signal amplitude > maximum | <ul style="list-style-type: none"> <li>▪ Rapid temperature change of ambient air temperature sensor</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ambient air temperature sensor failure</li> </ul>                                                                                                    | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) <ul style="list-style-type: none"> <li>▪ Check that a plausible value is displayed</li> </ul> </li> <li>▪ Refer to the electrical circuit diagrams and check ambient air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new ambient air temperature sensor as required</li> </ul>                       |
| P0071-23 | Ambient Air Temperature Sensor Circuit "A" Range /Performance - Signal stuck low           | <ul style="list-style-type: none"> <li>▪ Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)</li> <li>▪ Electric block heater applied and not detected</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ambient air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) - Engine Coolant Temperature (0xF405) - Boost Air Temperature - Raw physical value (0x0341) - Intake Air Temperature (0xF40F) - Engine Coolant Temperature #2 (0x0489). All sensors should be within 20 °C of each other</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>▪ Check and install new sensor that is biased higher or lower than the other sensors</li> </ul> |

|          |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0071-24 | Ambient Air Temperature Sensor Circuit "A" Range /Performance - Signal stuck high | <ul style="list-style-type: none"> <li>▪ Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)</li> <li>▪ Electric block heater applied and not detected</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ambient air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) - Engine Coolant Temperature (0xF405) - Boost Air Temperature - Raw physical value (0x0341) - Intake Air Temperature (0xF40F) - Engine Coolant Temperature #2 (0x0489). All sensors should be within 20 °C of each other</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>▪ Check and install new sensor that is biased higher or lower than the other sensors</li> </ul> |
| P0072-00 | Ambient Air Temperature Sensor Circuit "A" Low - No sub type information          | <ul style="list-style-type: none"> <li>▪ Ambient air temperature sensor circuit, short circuit to ground</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ambient air temperature sensor failure</li> </ul>                                                                                               | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check ambient air temperature sensor circuit for short circuit to ground</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new ambient air temperature sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                |
| P0073-00 | Ambient Air Temperature Sensor Circuit "A" High - No sub type information         | <ul style="list-style-type: none"> <li>▪ Ambient air temperature sensor circuit, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ambient air temperature sensor failure</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check ambient air temperature sensor circuit for short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new ambient air temperature sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| P007B-00 | Charge Air Cooler Temperature Sensor Circuit Range /Performance (Bank 1) - No sub type information | <ul style="list-style-type: none"> <li>■ Manifold absolute pressure and temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Manifold absolute pressure and temperature sensor failure</li> <li>■ Air charge coolant pump and control circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Air charge coolant pump relay failure</li> <li>■ Air charge coolant pump failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Refer to electrical circuit diagrams and check the air charge coolant pump and control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Refer to electrical circuit diagrams and check the air charge coolant pump for open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the relevant section of the workshop manual and check the air charge coolant pump for correct operation</li> <li>■ Check and install a new manifold absolute pressure and temperature sensor as required</li> <li>■ Check and install a new air charge coolant pump as required</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P007B-22 | Charge Air Cooler Temperature Sensor Circuit Range /Performance (Bank 1) - Signal                  | <ul style="list-style-type: none"> <li>■ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>■ Refer to electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|          |                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | amplitude > maximum                                                                         | <ul style="list-style-type: none"> <li>Manifold absolute pressure and temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Manifold absolute pressure and temperature sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new manifold absolute pressure and temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| P007B-23 | Charge Air Cooler Temperature Sensor Circuit Range /Performance (Bank 1) - Signal stuck low | <ul style="list-style-type: none"> <li>The engine control module measures a signal that remains low when transitions are expected</li> <li>Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)</li> <li>Electric block heater applied and not detected</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Manifold absolute pressure and temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Manifold absolute pressure and temperature sensor failure</li> <li><b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>Changeover valve - Fuel Fired Booster Heater (FFBH) system failure <ul style="list-style-type: none"> <li>This valve should only allow coolant into the engine if the engine coolant temperature is -25°C (-13°F) or below. When de-energized, the changeover valve connects the heater coolant circuit to the engine coolant circuit. When energized, the changeover valve isolates the heater coolant circuit from the engine coolant circuit. If this valve is de-energizing above -25°C (-13°F) it will be inadvertently allowing heated coolant into the engine and causing the DTC to set due to a mismatch between coolant temperature sensor 1 and all the other temperature sensors</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) - Engine Coolant Temperature (0xF405) - Boost Air Temperature - Raw physical value (0x0341) - Intake Air Temperature (0xF40F) - Engine Coolant Temperature #2 (0x0489). All sensors should be within 20 deg°C of each other</li> <li>Refer to electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new manifold absolute pressure and temperature sensor as required</li> <li>Check and install new sensor that is biased higher or lower than the other sensors</li> <li><b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> </ul> |

|          |                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>▪ Check and install new a changeover valve - Fuel Fired Booster Heater (FFBH) system</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P007B-24 | Charge Air Cooler Temperature Sensor Circuit Range /Performance (Bank 1) - Signal stuck high | <ul style="list-style-type: none"> <li>▪ The engine control module measures a signal that remains high when transitions are expected</li> <li>▪ Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)</li> <li>▪ Electric block heater applied and not detected</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuse failure</li> <li>▪ Manifold absolute pressure and temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Air charge coolant pump and control circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Air charge coolant pump relay failure</li> <li>▪ Air charge coolant pump failure</li> <li>▪ <b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>▪ Changeover valve - Fuel Fired Booster Heater (FFBH) system failure <ul style="list-style-type: none"> <li>▪ This valve should only allow coolant into the engine if the engine coolant temperature is -25°C (-13°F) or below. When de-energized, the changeover valve connects the heater coolant circuit to the engine coolant circuit. When energized, the changeover valve isolates the heater coolant circuit from the engine coolant circuit. If this valve is de-energizing above -25°C (-13°F) it will be inadvertently allowing heated coolant into</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>▪ Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) - Engine Coolant Temperature (0xF405) - Boost Air Temperature - Raw physical value (0x0341) - Intake Air Temperature (0xF40F) - Engine Coolant Temperature #2 (0x0489). All sensors should be within 20 deg°C of each other</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to electrical circuit diagrams and check the manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Refer to electrical circuit diagrams and check for fuse failure, install a new fuse as required</li> <li>▪ Check and install a new manifold absolute pressure and temperature sensor as required</li> <li>▪ Refer to electrical circuit diagrams and check the air charge coolant pump and control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Refer to electrical circuit diagrams and check the air charge coolant pump for open circuit, high resistance</li> </ul> |

|          |                                                                                                 |                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                 | <p>the engine and causing the DTC to set due to a mismatch between coolant temperature sensor 1 and all the other temperature sensors</p>                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Refer to the relevant section of the workshop manual and check the air charge coolant pump for correct operation</li> <li>Check and install a new air charge coolant pump as required</li> <li>Check and install new sensor that is biased higher or lower than the other sensors</li> <li><b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>Check and install new a changeover valve - Fuel Fired Booster Heater (FFBH) system</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P007C-00 | <p>Charge Air Cooler Temperature Sensor Circuit Low (Bank 1)<br/>- No sub type information</p>  | <ul style="list-style-type: none"> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new charge air temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                  |
| P007D-00 | <p>Charge Air Cooler Temperature Sensor Circuit High (Bank 1)<br/>- No sub type information</p> | <ul style="list-style-type: none"> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                 | <p>pins for damage and/or corrosion</p> <ul style="list-style-type: none"> <li>Check and install a new charge air temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                           |
| P007E-1F | Charge Air Cooler Temperature Sensor Circuit Intermittent /Erratic (Bank 1) - Circuit intermittent | <ul style="list-style-type: none"> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new charge air temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P007E-2F | Charge Air Cooler Temperature Sensor Circuit Intermittent /Erratic (Bank 1) - Signal erratic       | <ul style="list-style-type: none"> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new charge air temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0087-77 | Fuel Rail /System Pressure - Too Low - Commanded position not reachable                            | <div>  <b>NOTE:</b> </div> <div></div>                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Check for fuel leaks external of the fuel rail</li> <li>Check high pressure fuel pumps are not leaking</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                          |                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                          | <p><b>Monitor description.</b> To detect under pressure in the fuel rail. The high pressure fuel pump signal is below a threshold</p> <ul style="list-style-type: none"> <li>▪ Fuel leaking outside of the fuel system</li> <li>▪ Fuel leaking into the low pressure system</li> </ul>                                                                                                        | from the high pressure system into the low pressure system                                                                                                                                                                                                                                                                                                                                                                                        |
| P0087-84 | Fuel Rail /System Pressure - Too Low - Signal below allowable range      | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> The fuel rail pressure is below a threshold</p> <ul style="list-style-type: none"> <li>▪ Fuel rail pressure sensor circuit short circuit to ground</li> <li>▪ Fuel rail pressure sensor failure</li> </ul>                                               | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel rail pressure sensor as required</li> </ul>                                                                  |
| P0088-77 | Fuel Rail /System Pressure - Too High - Commanded position not reachable | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> To detect over pressure in the fuel rail. The high pressure fuel pump signal is above a threshold</p> <ul style="list-style-type: none"> <li>▪ Fuel leaking outside of the fuel system</li> <li>▪ Fuel leaking into the low pressure system</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check for fuel leaks external of the fuel rail</li> <li>▪ Check high pressure fuel pumps are not leaking from the high pressure system into the low pressure system</li> </ul>                                                                                                                                                                                                                           |
| P0088-85 | Fuel Rail /System Pressure - Too High - Signal above allowable range     | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> The fuel rail pressure is above a threshold</p> <ul style="list-style-type: none"> <li>▪ Fuel pressure relief valve stuck</li> <li>▪ Fuel rail pressure sensor circuit short circuit to power</li> <li>▪ Fuel rail pressure sensor failure</li> </ul>  | <ul style="list-style-type: none"> <li>▪ Check fuel pressure relief valve for correct operation</li> <li>▪ Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel rail pressure sensor as required</li> </ul> |

|          |                                                                            |                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0089-64 | Fuel Pressure Regulator Performance - Signal plausibility failure          |  <b>NOTE:</b><br><br><b>Monitor description.</b> The command signal to the fuel pump driver module is above a threshold     | <ul style="list-style-type: none"> <li>Check for fuel system leakage from the fuel tank and lines both internal and external of the fuel tank</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new low pressure fuel pump as required</li> </ul>                                                                         |
| P0089-92 | Fuel Pressure Regulator Performance - Performance or incorrect operation   |  <b>NOTE:</b><br><br><b>Monitor description.</b> The command signal to the fuel pump driver module is below a threshold     | <ul style="list-style-type: none"> <li>Check for fuel system leakage from the fuel tank and lines both internal and external of the fuel tank</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new low pressure fuel pump as required</li> </ul>                                                                         |
| P008A-07 | Low Pressure Fuel System Pressure - Too Low - Mechanical Failures          |  <b>NOTE:</b><br><br><b>Monitor description.</b> The monitor checks the low pressure fuel is not below the expected range | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel low pressure sensor as required</li> </ul>                                                                      |
| P008A-84 | Low Pressure Fuel System Pressure - Too Low - Signal below allowable range |  <b>NOTE:</b><br><br><b>Monitor description.</b> Low pressure fuel signal is below the threshold                          | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check for fuel system leakage from the fuel tank and lines both internal and external of the fuel tank</li> </ul> |

|          |                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new low pressure fuel pump as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |
| P008B-07 | Low Pressure Fuel System Pressure - Too High - Mechanical Failures          | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The monitor checks the low pressure fuel is not above the expected range</p> </div> <ul style="list-style-type: none"> <li>Fuel low pressure sensor circuit short circuit to ground, high resistance</li> <li>Fuel low pressure sensor failure</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for short circuit to ground, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel low pressure sensor as required</li> </ul>                                                                                                                                                                                                                                                  |
| P008B-85 | Low Pressure Fuel System Pressure - Too High - Signal above allowable range | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Low pressure fuel signal is above the threshold</p> </div> <ul style="list-style-type: none"> <li>Fuel low pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel system leakage from the fuel tank and lines</li> <li>Low pressure fuel pump failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel low pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check for fuel system leakage from the fuel tank and lines both internal and external of the fuel tank</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new low pressure fuel pump as required</li> </ul> |
| P0095-00 | Intake Air Temperature Sensor 2 Circuit (Bank 1) - No sub type information  | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |

|          |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P0096-00 | Intake Air Temperature Sensor 2 Circuit Range /Performance (Bank 1) - No sub type information | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Charge air temperature sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install a new charge air temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                        |
| P0096-23 | Intake Air Temperature Sensor 2 Circuit Range /Performance (Bank 1) - Signal stuck low        | <ul style="list-style-type: none"> <li>The engine control module measures a signal that remains low when transitions are expected</li> <li>Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)</li> <li>Electric block heater applied and not detected</li> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor failure</li> <li><b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>Changeover valve - Fuel Fired Booster Heater (FFBH) system failure <ul style="list-style-type: none"> <li>This valve should only allow coolant into the engine if the engine coolant temperature is -25°C (-13°F) or below. When de-energized, the changeover valve connects the heater</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) - Engine Coolant Temperature (0xF405) - Boost Air Temperature - Raw physical value (0x0341) - Intake Air Temperature (0xF40F) - Engine Coolant Temperature #2 (0x0489). All sensors should be within 20 deg°C of each other</li> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new charge air temperature sensor as required</li> </ul> |

|          |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                         | <p>coolant circuit to the engine coolant circuit. When energized, the changeover valve isolates the heater coolant circuit from the engine coolant circuit. If this valve is de-energizing above -25°C (-13°F) it will be inadvertently allowing heated coolant into the engine and causing the DTC to set due to a mismatch between coolant temperature sensor 1 and all the other temperature sensors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Check and install new sensor that is biased higher or lower than the other sensors</li> <li><b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>Check and install new a changeover valve - Fuel Fired Booster Heater (FFBH) system</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| P0096-24 | Intake Air Temperature Sensor 2 Circuit Range /Performance (Bank 1) - Signal stuck high | <ul style="list-style-type: none"> <li>The engine control module measures a signal that remains high when transitions are expected</li> <li>Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)</li> <li>Electric block heater applied and not detected</li> <li>Fuse failure</li> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor failure</li> <li><b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>Changeover valve - Fuel Fired Booster Heater (FFBH) system failure <ul style="list-style-type: none"> <li>This valve should only allow coolant into the engine if the engine coolant temperature is -25°C (-13°F) or below. When de-energized, the changeover valve connects the heater coolant circuit to the engine coolant circuit. When energized, the changeover valve isolates the heater coolant circuit from the engine coolant circuit. If this valve is</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) - Engine Coolant Temperature (0xF405) - Boost Air Temperature - Raw physical value (0x0341) - Intake Air Temperature (0xF40F) - Engine Coolant Temperature #2 (0x0489). All sensors should be within 20 deg°C of each other</li> <li>Refer to electrical circuit diagrams and check for fuse failure, install a new fuse as required</li> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new charge air temperature sensor as required</li> <li><b>Additional checks to carry out ONLY for Range Rover</b></li> </ul> |

|          |                                                                                 |                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                 | <p>de-energizing above -25°C (-13°F) it will be inadvertently allowing heated coolant into the engine and causing the DTC to set due to a mismatch between coolant temperature sensor 1 and all the other temperature sensors</p>                                                                               | <p><b>/ L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></p> <ul style="list-style-type: none"> <li>Check and install new a changeover valve - Fuel Fired Booster Heater (FFBH) system</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                 |
| P0097-00 | Intake Air Temperature Sensor 2 Circuit Low (Bank 1) - No sub type information  | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Charge air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install a new charge air temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0098-00 | Intake Air Temperature Sensor 2 Circuit High (Bank 1) - No sub type information | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Charge air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install a new charge air temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0099-1F | Intake Air Temperature Sensor 2                                                 | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|          |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Circuit Intermittent /Erratic (Bank 1) - Circuit intermittent                                     | <ul style="list-style-type: none"> <li>Charge air temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Charge air temperature sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the charge air temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install a new charge air temperature sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                     |
| P00B6-00 | Radiator Coolant Temperature / Engine Coolant Temperature Correlation - No sub type information   | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                             |
| P00BC-00 | Mass or Volume Air Flow A Circuit Range /Performance - Air Flow Too Low - No sub type information | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Rationality. Mass air flow sensor flow to low </div> <ul style="list-style-type: none"> <li>Blocked air filter</li> <li>Other related DTCs</li> <li>Air induction system, incorrectly installed components, loose hose clips or air leakage</li> <li>Mass air flow sensor contamination by oil or debris</li> <li>Electric throttle blade contamination by oil or debris</li> <li>Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Throttle adaption failure</li> </ul> | <ul style="list-style-type: none"> <li>Check air filter is serviceable</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and refer to the relevant DTC index</li> <li>Refer to the relevant sections of the workshop manual and check the induction system for correctly installed components, loose hose clips or air leakage</li> <li>Check mass air flow sensor for contamination by oil or debris clean as required</li> <li>Check electric throttle blade for contamination by oil or debris clean as required</li> <li>Refer to the electrical circuit diagrams and check</li> </ul> |

|          |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment carry out throttle adaption routine</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| P00BD-00 | <p>Mass or Volume Air Flow A Circuit Range /Performance - Air Flow Too High - No sub type information</p> | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b><br/>Rationality. Mass air flow sensor flow excessive</p> </div> <ul style="list-style-type: none"> <li>▪ Blocked air filter</li> <li>▪ Other related DTCs</li> <li>▪ Air induction system, incorrectly installed components, loose hose clips or air leakage</li> <li>▪ Mass air flow sensor contamination by oil or debris</li> <li>▪ Electric throttle blade contamination by oil or debris</li> <li>▪ Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Throttle adaption failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check air filter is serviceable</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and refer to the relevant DTC index</li> <li>▪ Refer to the relevant sections of the workshop manual and check the induction system for correctly installed components, loose hose clips or air leakage</li> <li>▪ Check mass air flow sensor for contamination by oil or debris clean as required</li> <li>▪ Check electric throttle blade for contamination by oil or debris clean as required</li> <li>▪ Refer to the electrical circuit diagrams and check mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment carry out throttle adaption routine</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored</li> </ul> |

|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DTCs using the 'Diagnosis Menu' tab and retest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| P00BE-00 | Mass or Volume Air Flow B Circuit Range /Performance - Air Flow Too Low - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Rationality. Mass air flow sensor flow to low </div> <ul style="list-style-type: none"> <li>Blocked air filter</li> <li>Other related DTCs</li> <li>Air induction system, incorrectly installed components, loose hose clips or air leakage</li> <li>Mass air flow sensor contamination by oil or debris</li> <li>Electric throttle blade contamination by oil or debris</li> <li>Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Throttle adaption failure</li> </ul> | <ul style="list-style-type: none"> <li>Check air filter is serviceable</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and refer to the relevant DTC index</li> <li>Refer to the relevant sections of the workshop manual and check the induction system for correctly installed components, loose hose clips or air leakage</li> <li>Check mass air flow sensor for contamination by oil or debris clean as required</li> <li>Check electric throttle blade for contamination by oil or debris clean as required</li> <li>Refer to the electrical circuit diagrams and check mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment carry out throttle adaption routine</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P00BF-00 | Mass or Volume Air Flow B Circuit Range /Performance - Air Flow Too High - No sub type information | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Rationality. Mass air flow sensor flow excessive </div> <ul style="list-style-type: none"> <li>Blocked air filter</li> <li>Other related DTCs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Check air filter is serviceable</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check the engine control module for related DTCs and refer to the relevant DTC index</li> <li>Refer to the relevant sections of the workshop manual and check the induction system for</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|          |                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                      | <ul style="list-style-type: none"> <li>▪ Air induction system, incorrectly installed components, loose hose clips or air leakage</li> <li>▪ Mass air flow sensor contamination by oil or debris</li> <li>▪ Electric throttle blade contamination by oil or debris</li> <li>▪ Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Throttle adaption failure</li> </ul>                                             | <p>correctly installed components, loose hose clips or air leakage</p> <ul style="list-style-type: none"> <li>▪ Check mass air flow sensor for contamination by oil or debris clean as required</li> <li>▪ Check electric throttle blade for contamination by oil or debris clean as required</li> <li>▪ Refer to the electrical circuit diagrams and check mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment carry out throttle adaption routine</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P00C6-00 | Fuel Rail Pressure Too Low - Engine Cranking - No sub type information                               | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The fuel rail pressure is checked against a threshold after a number of engine rotations from the start of cranking. A DTC is set if the fuel rail pressure is below the threshold</p> </div> <ul style="list-style-type: none"> <li>▪ Fuel rail has been removed</li> <li>▪ High pressure fuel pump mechanical failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check the fuel rail pressure during cranking. Clear the DTC and attempt another start</li> <li>▪ Check and install new high pressure fuel pump as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| P00CE-21 | Intake Air Temperature Measurement System - Multiple Sensor Correlation - Signal amplitude < minimum | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Harness failure - Wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|          |                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                 |
| P00CE-22 | Intake Air Temperature Measurement System - Multiple Sensor Correlation - Signal amplitude > maximum | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                            |
| P00D1-00 | HO2S Heater Control Circuit Range /Performance (Bank 1 Sensor 1) - No sub type information           | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> The value of the correction factor of the heated oxygen sensor calibration resistor is checked against a threshold. A DTC is set if the value is above the threshold</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P00D1-29 | HO2S Heater Control Circuit Range /Performance (Bank 1 Sensor 1) - Signal invalid                    | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> The engine control module monitors the ceramic temperature of the heated oxygen sensor nernst cell to ensure it is sufficiently heated after engine start</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                         | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen</li> </ul>                                   |

|          |                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                             | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sensors (HO2S), Diagnosis and Testing - pinpoint test A                                                                                                                                                                                                                                                                                                                                                                                                                           |
| P00D1-64 | HO2S Heater Control Circuit Range /Performance (Bank 1 Sensor 1) - Signal invalid           | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The engine control module monitors the ceramic temperature of the heated oxygen sensor nernst cell to ensure it is sufficiently heated after engine start</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>              | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P00D3-00 | HO2S Heater Control Circuit Range /Performance (Bank 2 Sensor 1) - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The value of the correction factor of the heated oxygen sensor calibration resistor is checked against a threshold. A DTC is set if the value is above the threshold</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P00D3-04 | HO2S Heater Control Circuit Range /Performance (Bank 2 Sensor 1) - System Internal Failures | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The value of the correction factor of the heated oxygen sensor calibration resistor is checked against a threshold. A DTC is set if the value is above the threshold</p> </div>                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                        |

|          |                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| P00D3-64 | HO2S Heater Control Circuit Range /Performance (Bank 2 Sensor 1) - Signal plausibility failure | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> The engine control module monitors the ceramic temperature of the heated oxygen sensor nernst cell to ensure it is sufficiently heated after engine start</p> </div> <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul>        | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater control circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                                                                                                                                              |
| P0100-00 | Mass or Volume Air Flow Sensor "A" Circuit - No sub type information                           | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> Engine control module monitors the input frequency signal from the sensor and sets the DTC if there is electrical failure</p> </div> <ul style="list-style-type: none"> <li>▪ Mass air flow sensor connector is not correctly latched</li> <li>▪ Power feed to mass air flow sensor, open circuit</li> <li>▪ Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Mass air flow sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Mass Air Flow Sensor Frequency - (0x033D)</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Air Flow Rate from Mass Air Flow Sensor - (0xF410)</li> <li>▪ Check mass air flow sensor connector is correctly latched</li> <li>▪ Refer to electrical circuit diagrams and check power feed to mass air flow sensor for open circuit</li> <li>▪ Refer to electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul> |

|          |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check and install a new mass air flow sensor as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P0101-00 | Mass or Volume Air Flow Sensor "A" Circuit Range /Performance - No sub type information | <ul style="list-style-type: none"> <li>▪ Mass air flow sensor contamination by oil or debris</li> <li>▪ Mass air flow sensor connector is not correctly latched</li> <li>▪ Power feed to mass air flow sensor, open circuit</li> <li>▪ Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Mass air flow sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Mass Air Flow Sensor Frequency - (0x033D)</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Air Flow Rate from Mass Air Flow Sensor - (0xF410) <ul style="list-style-type: none"> <li>▪ Verify signal is operational when engine is idling</li> </ul> </li> <li>▪ Check mass air flow sensor for contamination by oil or debris clean as required</li> <li>▪ Check mass air flow sensor connector is correctly latched</li> <li>▪ Refer to electrical circuit diagrams and check power feed to mass air flow sensor for open circuit</li> <li>▪ Refer to electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check and install a new mass air flow sensor as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |

|          |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0102-00 | Mass or Volume Air Flow Sensor "A" Circuit Low - No sub type information       | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Engine control module monitors the input frequency signal from the sensor and sets the DTC if there is electrical failure</p> </div> <ul style="list-style-type: none"> <li>Mass air flow sensor connector is not correctly latched</li> <li>Power feed to mass air flow sensor, open circuit</li> <li>Mass air flow sensor circuit, short circuit to ground</li> <li>Mass air flow sensor failure</li> </ul>   | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Mass Air Flow Sensor Frequency - (0x033D)</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Air Flow Rate from Mass Air Flow Sensor - (0xF410)</li> <li>Check mass air flow sensor connector is correctly latched</li> <li>Refer to electrical circuit diagrams and check power feed to mass air flow sensor for open circuit</li> <li>Refer to electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new mass air flow sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0103-85 | Mass or Volume Air Flow Sensor "A" Circuit High - Signal above allowable range | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Engine control module monitors the input frequency signal from the sensor and sets the DTC if there is electrical failure</p> </div> <ul style="list-style-type: none"> <li>Mass air flow sensor connector is not correctly latched</li> <li>Power feed to mass air flow sensor, open circuit</li> <li>Mass air flow sensor circuit, short circuit to power</li> <li>Mass air flow sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Mass Air Flow Sensor Frequency - (0x033D)</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Air Flow Rate from Mass Air Flow Sensor - (0xF410)</li> <li>Check mass air flow sensor connector is correctly latched</li> <li>Refer to electrical circuit diagrams and check power feed to mass air flow sensor for open circuit</li> <li>Refer to electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to power</li> </ul>                                                                                                                                                                                                                                                                                                    |

|          |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new mass air flow sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| P0106-00 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit Range /Performance - No sub type information | <ul style="list-style-type: none"> <li>Manifold absolute pressure and temperature sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure and temperature sensor failure</li> <li>Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> <li>Check throttle blade for carbon obstruction</li> <li>Check and install new manifold absolute pressure and temperature sensor as required</li> <li>Check catalytic converter for damage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0106-21 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit                                              | <ul style="list-style-type: none"> <li>Manifold absolute pressure and temperature sensor circuit, for short circuit to ground</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> </ul>                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|          |                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p>Range /Performance<br/>- Signal amplitude &lt; minimum</p>                                                                | <ul style="list-style-type: none"> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure and temperature sensor failure</li> <li>Damaged catalytic converter</li> </ul>                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> <li>Check throttle blade for carbon obstruction</li> <li>Check and install new manifold absolute pressure and temperature sensor as required</li> <li>Check catalytic converter for damage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0106-22 | <p>Manifold Absolute Pressure /Barometric Pressure Sensor Circuit Range /Performance<br/>- Signal amplitude &gt; maximum</p> | <ul style="list-style-type: none"> <li>Manifold absolute pressure and temperature sensor circuit, for short circuit to power</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure and temperature sensor failure</li> <li>Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to power</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> <li>Check throttle blade for carbon obstruction</li> <li>Check and install new manifold absolute pressure</li> </ul>                                                                    |

|          |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>and temperature sensor as required</p> <ul style="list-style-type: none"> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| P0106-23 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit Range /Performance - Signal stuck low  | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to ground</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure and temperature sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0106-24 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit Range /Performance - Signal stuck high | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to power</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> </ul>                                                                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to power</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|          |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                    | <ul style="list-style-type: none"> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul>                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure and temperature sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                      |
| P0106-29 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit Range /Performance - Signal invalid | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure and temperature sensor as required</li> </ul> |

|          |                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| P0106-84 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit Range /Performance - Signal below allowable range | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to ground</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure and temperature sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0106-85 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit Range /Performance - Signal above allowable range | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to power</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> </ul>                                                                                                              | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to power</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|          |                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                              | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul>                                                                                                                                                                                                                                                                                                                                                        | <p>position voltage at ignition on</p> <ul style="list-style-type: none"> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure and temperature sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                   |
| P0107-00 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit Low - No sub type information | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure and temperature sensor as required</li> <li>▪ Check catalytic converter for damage</li> </ul> |

|          |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| P0108-00 | Manifold Absolute Pressure /Barometric Pressure Sensor Circuit High - No sub type information | <ul style="list-style-type: none"> <li>Manifold absolute pressure and temperature sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure and temperature sensor failure</li> <li>Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> <li>Check throttle blade for carbon obstruction</li> <li>Check and install new manifold absolute pressure and temperature sensor as required</li> <li>Check catalytic converter for damage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P010A-00 | Mass or Volume Air Flow Sensor "B" Circuit - No sub type information                          | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Engine control module monitors the input frequency signal from the sensor and sets the DTC if there is electrical failure</p> </div> <ul style="list-style-type: none"> <li>Mass air flow sensor connector is not correctly latched</li> </ul>                                                                                           | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Mass Air Flow Sensor Frequency - (0x033D)</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Air Flow Rate from Mass Air Flow Sensor - (0xF410)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|          |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                         | <ul style="list-style-type: none"> <li>▪ Power feed to mass air flow sensor, open circuit</li> <li>▪ Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Mass air flow sensor failure</li> </ul>                                                                                                                                   | <ul style="list-style-type: none"> <li>▪ Check mass air flow sensor connector is correctly latched</li> <li>▪ Refer to electrical circuit diagrams and check power feed to mass air flow sensor for open circuit</li> <li>▪ Refer to electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check and install a new mass air flow sensor as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                       |
| P010B-00 | Mass or Volume Air Flow Sensor "B" Circuit Range /Performance - No sub type information | <ul style="list-style-type: none"> <li>▪ Mass air flow sensor contamination by oil or debris</li> <li>▪ Mass air flow sensor connector is not correctly latched</li> <li>▪ Power feed to mass air flow sensor, open circuit</li> <li>▪ Mass air flow sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Mass air flow sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Mass Air Flow Sensor Frequency - (0x033D)</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Air Flow Rate from Mass Air Flow Sensor - (0xF410) <ul style="list-style-type: none"> <li>▪ Verify signal is operational when engine is idling</li> </ul> </li> <li>▪ Check mass air flow sensor for contamination by oil or debris clean as required</li> <li>▪ Check mass air flow sensor connector is correctly latched</li> <li>▪ Refer to electrical circuit diagrams and check power feed to mass air flow sensor for open circuit</li> <li>▪ Refer to electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul> |

|          |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new mass air flow sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                             |
| P010C-00 | Mass or Volume Air Flow Sensor "B" Circuit Low - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Engine control module monitors the input frequency signal from the sensor and sets the DTC if there is electrical failure</p> </div> <ul style="list-style-type: none"> <li>Mass air flow sensor circuit, short circuit to ground</li> <li>Mass air flow sensor failure</li> </ul>  | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Mass Air Flow Sensor Frequency - (0x033D)</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Air Flow Rate from Mass Air Flow Sensor - (0xF410)</li> <li>Refer to electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to ground</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new mass air flow sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P010D-00 | Mass or Volume Air Flow Sensor "B" Circuit High - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Engine control module monitors the input frequency signal from the sensor and sets the DTC if there is electrical failure</p> </div> <ul style="list-style-type: none"> <li>Mass air flow sensor circuit, short circuit to power</li> <li>Mass air flow sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Mass Air Flow Sensor Frequency - (0x033D)</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Air Flow Rate from Mass Air Flow Sensor - (0xF410)</li> <li>Refer to electrical circuit diagrams and check the mass air flow sensor circuit for short circuit to power</li> </ul>                                                                                                                                                                                                                                                                                                    |

|          |                                                                                                  |                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                  |                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check and install a new mass air flow sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                      |
| P0111-21 | Intake Air Temperature Sensor 1 Circuit Range /Performance (Bank 1) - Signal amplitude < minimum | <ul style="list-style-type: none"> <li>The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul>     | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Charge Air Temperature Voltage - (0x03EE) &amp; Intake Air Temperature Sensor Voltage - (0x1279)</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0111-22 | Intake Air Temperature Sensor 1 Circuit Range /Performance (Bank 1) - Signal amplitude > maximum | <ul style="list-style-type: none"> <li>The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Charge Air Temperature Voltage - (0x03EE) &amp; Intake Air Temperature Sensor Voltage - (0x1279)</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0111-   | Intake Air                                                                                       |                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                                            |                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23       | Temperature Sensor 1<br>Circuit Range /Performance (Bank 1) - Signal stuck low             | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul>                                                                            | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                       |
| P0111-24 | Intake Air Temperature Sensor 1<br>Circuit Range /Performance (Bank 1) - Signal stuck high | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul>                                                                            | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                       |
| P0111-29 | Intake Air Temperature Sensor 1<br>Circuit Range /Performance (Bank 1) - Signal invalid    | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul>                                                                            | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                       |
| P0112-00 | Intake Air Temperature Sensor 1<br>Circuit Low (Bank 1) - No sub type information          | <ul style="list-style-type: none"> <li>Intake air temperature sensor circuit, short circuit to ground</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Intake air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check intake air temperature sensor circuit for short circuit to ground</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                          |                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Check and install new intake air temperature sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| P0113-00 | Intake Air Temperature Sensor 1 Circuit High (Bank 1) - No sub type information          | <ul style="list-style-type: none"> <li>Intake air temperature sensor circuit open circuit, short circuit to power</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Intake air temperature sensor failure</li> </ul>                                   | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check intake air temperature sensor circuit for open circuit, short circuit to power</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new intake air temperature sensor as required</li> </ul>                          |
| P0114-00 | Intake Air Temperature Sensor 1 Intermittent /Erratic (Bank 1) - No sub type information | <ul style="list-style-type: none"> <li>Intake air temperature sensor circuit open circuit, short circuit to power, short circuit to ground</li> <li>Connector is disconnected or loose, connector pin is backed out, connector pin corrosion</li> <li>Intake air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check intake air temperature sensor circuit for open circuit, short circuit to power, short circuit to ground</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new intake air temperature sensor as required</li> </ul> |
| P0114-1F | Intake Air Temperature Sensor 1 Intermittent /Erratic (Bank 1) - Circuit intermittent    | <ul style="list-style-type: none"> <li>Intake air temperature sensor circuit open circuit, short circuit to power, short circuit to ground</li> <li>Connector is disconnected or loose, connector pin is backed out, connector pin corrosion</li> <li>Intake air temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check intake air temperature sensor circuit for open circuit, short circuit to power, short circuit to ground</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored</li> </ul>                                                                                                                         |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                          |                                                                                                                                                                                                                                                                                                                        | <p>DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>Check and install new intake air temperature sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                        |
| P0115-00 | Engine Coolant Temperature Sensor 1 Circuit - No sub type information                    | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Engine coolant temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine coolant temperature sensor as required</li> </ul> |
| P0115-1F | Engine Coolant Temperature Sensor 1 Circuit - Circuit intermittent                       | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Engine coolant temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine coolant temperature sensor as required</li> </ul> |
| P0116-00 | Engine Coolant Temperature Sensor 1 Circuit Range /Performance - No sub type information | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                    | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit</li> </ul>                                                                                                                                                                                                                                                                                                    |

|          |                                                                                             |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                             | <ul style="list-style-type: none"> <li>Engine coolant temperature sensor failure</li> </ul>                                                                                                                                                                                                                            | <p>for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine coolant temperature sensor as required</li> </ul>                                                                                                                                                                                             |
| P0116-21 | Engine Coolant Temperature Sensor 1 Circuit Range /Performance - Signal amplitude < minimum | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Engine coolant temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine coolant temperature sensor as required</li> </ul> |
| P0116-22 | Engine Coolant Temperature Sensor 1 Circuit Range /Performance - Signal amplitude > maximum | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Engine coolant temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine coolant temperature sensor as required</li> </ul> |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0116-23 | Engine Coolant Temperature Sensor 1<br>Circuit Range /Performance<br>- Signal stuck low  | <ul style="list-style-type: none"> <li>■ The engine control module measures a signal that remains low when transitions are expected</li> <li>■ Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)</li> <li>■ Electric block heater applied and not detected</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Engine coolant temperature sensor failure</li> <li>■ <b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>■ Changeover valve - Fuel Fired Booster Heater (FFBH) system failure <ul style="list-style-type: none"> <li>■ This valve should only allow coolant into the engine if the engine coolant temperature is -25°C (-13°F) or below. When de-energized, the changeover valve connects the heater coolant circuit to the engine coolant circuit. When energized, the changeover valve isolates the heater coolant circuit from the engine coolant circuit. If this valve is de-energizing above -25°C (-13°F) it will be inadvertently allowing heated coolant into the engine and causing the DTC to set due to a mismatch between coolant temperature sensor 1 and all the other temperature sensors</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>■ Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) - Engine Coolant Temperature (0xF405) - Boost Air Temperature - Raw physical value (0x0341) - Intake Air Temperature (0xF40F) - Engine Coolant Temperature #2 (0x0489). All sensors should be within 20 deg°C of each other</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>■ Check and install new sensor that is biased higher or lower than the other sensors</li> <li>■ <b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>■ Check and install new a changeover valve - Fuel Fired Booster Heater (FFBH) system</li> </ul> |
| P0116-24 | Engine Coolant Temperature Sensor 1<br>Circuit Range /Performance<br>- Signal stuck high | <ul style="list-style-type: none"> <li>■ The engine control module measures a signal that remains high when transitions are expected</li> <li>■ Battery disconnection resulting in errors in engine off time (short soaks may look like long soaks)</li> <li>■ Electric block heater applied and not detected</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Engine coolant temperature sensor failure</li> <li>■ <b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>■ Leave vehicle turned off for a minimum of 8 hours and allow to soak to a stable temperature. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Ambient Air Temperature - (0xF446) - Engine Coolant Temperature (0xF405) - Boost Air Temperature - Raw physical value (0x0341) - Intake Air Temperature (0xF40F) - Engine Coolant Temperature #2 (0x0489).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|          |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                            | <p><b>equipped with Fuel Fired Booster Heater (FFBH) system</b></p> <ul style="list-style-type: none"> <li>Changeover valve - Fuel Fired Booster Heater (FFBH) system failure <ul style="list-style-type: none"> <li>This valve should only allow coolant into the engine if the engine coolant temperature is -25°C (-13°F) or below. When de-energized, the changeover valve connects the heater coolant circuit to the engine coolant circuit. When energized, the changeover valve isolates the heater coolant circuit from the engine coolant circuit. If this valve is de-energizing above -25°C (-13°F) it will be inadvertently allowing heated coolant into the engine and causing the DTC to set due to a mismatch between coolant temperature sensor 1 and all the other temperature sensors</li> </ul> </li> </ul> | <p>All sensors should be within 20 deg°C of each other</p> <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Check and install new sensor that is biased higher or lower than the other sensors</li> <li><b>Additional checks to carry out ONLY for Range Rover / L405 and Range Rover Sport / L494 vehicles equipped with Fuel Fired Booster Heater (FFBH) system</b></li> <li>Check and install new a changeover valve - Fuel Fired Booster Heater (FFBH) system</li> </ul> |
| P0117-00 | Engine Coolant Temperature Sensor 1 Circuit Low - No sub type information  | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Engine coolant temperature sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine coolant temperature sensor as required</li> </ul>                                                                                                                          |
| P0118-00 | Engine Coolant Temperature Sensor 1 Circuit High - No sub type information | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Engine coolant temperature sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to ground, short circuit to power,</li> </ul>                                                                                                                                                                                                                                                                                                                                                                        |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                          |                                                                                                                                                                                                                                                                                                                        | <p>open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine coolant temperature sensor as required</li> </ul>                                                                                                                                                                                                                                                  |
| P0119-1F | Engine Coolant Temperature Sensor 1 Circuit Intermittent /Erratic - Circuit intermittent | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Engine coolant temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine coolant temperature sensor as required</li> </ul> |
| P011B-23 | Engine Coolant Temperature / Intake Air Temperature Correlation - Signal stuck low       | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor failure</li> <li>Intake air temperature sensor failure</li> </ul>                                                                                    | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                               |
| P011B-24 | Engine Coolant Temperature / Intake Air Temperature Correlation - Signal stuck high      | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Engine coolant temperature sensor failure</li> <li>Intake air temperature sensor failure</li> </ul>                                                                                    | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                  |

|          |                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P0121-00 | Throttle/Pedal Position Sensor A Circuit Range /Performance<br>- No sub type information  | <ul style="list-style-type: none"> <li>Electrical Cause <ul style="list-style-type: none"> <li>Yes</li> </ul> </li> <li>Mechanical Cause <ul style="list-style-type: none"> <li>No</li> </ul> </li> <li>Control module cavity <ul style="list-style-type: none"> <li>Circuit reference I_A_TVA1</li> </ul> </li> <li>Monitor Description <ul style="list-style-type: none"> <li>Throttle position sensor 1 does not align to position sensor 2 and is also not aligned to a theoretical throttle angle calculated from air charge / load. Throttle position sensor 1 is electrically within valid range but irrational to operating conditions</li> </ul> </li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Electric throttle - Wiring integrity <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging Strategy <ul style="list-style-type: none"> <li>Ensure all throttle position circuit fault have been rectified prior to investigating this fault</li> <li>Engine running at idle for 5 seconds, also perform a short (less than 10 minute) varied drive cycle to ensure fault has been rectified, ensure light and high engine loads are included</li> </ul> </li> <li>Using the Jaguar Land Rover approved diagnostic equipment, with ignition on but engine off, check electric throttle position sensor signal 1 is aligned to electric throttle position sensor signal 2</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check the electric throttle circuit for <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0121-04 | Throttle/Pedal Position Sensor A Circuit Range /Performance<br>- System Internal Failures | <ul style="list-style-type: none"> <li>Electrical Cause <ul style="list-style-type: none"> <li>Yes</li> </ul> </li> <li>Mechanical Cause <ul style="list-style-type: none"> <li>No</li> </ul> </li> <li>Control module cavity <ul style="list-style-type: none"> <li>Not applicable</li> </ul> </li> <li>Monitor Description</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging Strategy <ul style="list-style-type: none"> <li>During electric throttle adaption process at ignition ON engine OFF the amplifier is checked</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|          |                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                              | <ul style="list-style-type: none"> <li>Amplified signal is out of range of expected 4 X amplification from raw TPS1 input signal, Throttle amplifier circuit has failed within the engine control module and the engine control module must be replaced</li> <li>Other related DTCs</li> <li>Engine control module failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Check and install a new engine control module, only when diagnosed as failed</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                        |
| P0122-00 | Throttle/Pedal Position Sensor A Circuit Low - No sub type information       | <ul style="list-style-type: none"> <li>Electrical Cause <ul style="list-style-type: none"> <li>Yes</li> </ul> </li> <li>Mechanical Cause <ul style="list-style-type: none"> <li>No</li> </ul> </li> <li>Control module cavity <ul style="list-style-type: none"> <li>Circuit reference I_A_TVA1 (engine control module)</li> <li>Circuit reference POT 1 (electric throttle)</li> </ul> </li> <li>Monitor Description <ul style="list-style-type: none"> <li>Throttle position sensor 1 is below valid electrical range</li> </ul> </li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Electric throttle - Wiring integrity <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging Strategy <ul style="list-style-type: none"> <li>Ignition On, Engine greater than 1200 rpm for 5 seconds</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check the electric throttle circuit for <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0123-85 | Throttle/Pedal Position Sensor A Circuit High - Signal above allowable range | <ul style="list-style-type: none"> <li>Electrical Cause <ul style="list-style-type: none"> <li>Yes</li> </ul> </li> <li>Mechanical Cause <ul style="list-style-type: none"> <li>No</li> </ul> </li> <li>Control module cavity <ul style="list-style-type: none"> <li>Circuit reference I_A_TVA1 (engine control module)</li> <li>Circuit reference POT 1 (electric throttle)</li> </ul> </li> <li>Monitor Description <ul style="list-style-type: none"> <li>Throttle position sensor 1 is above valid electrical range</li> </ul> </li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Electric throttle - Wiring integrity <ul style="list-style-type: none"> <li>Short circuit to ground</li> </ul> </li> </ul>                                                                                | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging Strategy <ul style="list-style-type: none"> <li>Ignition On, Engine greater than 1200 rpm for 5 seconds</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check the electric throttle circuit for <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> </ul>                                                                                                                                    |

|          |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                     | <ul style="list-style-type: none"> <li>▪ Short circuit to power</li> <li>▪ Open circuit</li> <li>▪ High resistance</li> </ul>                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      |
| P0126-00 | Insufficient Coolant Temp For Stable Operation - No sub type information                            | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Engine coolant temperature sensor failure</li> </ul>                                                                                       | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new engine coolant temperature sensor as required</li> </ul>             |
| P0128-00 | Coolant Thermostat (Coolant Temp Below Thermostat Regulating Temperature) - No sub type information | <ul style="list-style-type: none"> <li>▪ Thermostat stuck open or partially open</li> <li>▪ Thermostat failure</li> </ul>                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>▪ Check and install new thermostat</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| P012A-23 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit - Signal stuck low                         | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure sensor circuit, for short circuit to ground</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure sensor failure</li> <li>▪ Damaged catalytic converter</li> <li>▪ Blocked exhaust system</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> </ul> |

|          |                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>▪ Check and install new manifold absolute pressure sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Check exhaust system for blockage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| P012B-21 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit Range /Performance - Signal amplitude < minimum | <ul style="list-style-type: none"> <li>▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low</li> <li>▪ Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure sensor failure</li> <li>▪ Damaged catalytic converter</li> <li>▪ Blocked exhaust system</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Check exhaust system for blockage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P012B-22 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit Range                                           | <ul style="list-style-type: none"> <li>▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|          |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | /Performance<br>- Signal amplitude > maximum                                                      | <ul style="list-style-type: none"> <li>Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure sensor failure</li> <li>Damaged catalytic converter</li> <li>Blocked exhaust system</li> </ul>                                                                                                     | circuit to power, open circuit, high resistance <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> <li>Check throttle blade for carbon obstruction</li> <li>Check and install new manifold absolute pressure sensor as required</li> <li>Check catalytic converter for damage</li> <li>Check exhaust system for blockage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P012B-23 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit Range /Performance<br>- Signal stuck low | <ul style="list-style-type: none"> <li>The engine control module measures a signal that remains low when transitions are expected</li> <li>Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure sensor failure</li> <li>Damaged catalytic converter</li> <li>Blocked exhaust system</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> </ul>                                                                                                                                                                                                                      |

|          |                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Check exhaust system for blockage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| P012B-24 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit Range /Performance - Signal stuck high | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure sensor circuit, for short circuit to power</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure sensor failure</li> <li>▪ Damaged catalytic converter</li> <li>▪ Blocked exhaust system</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to power</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Check exhaust system for blockage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P012B-29 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit Range /Performance - Signal invalid    | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> </ul>                                                                                                                                                        | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|          |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                            | <ul style="list-style-type: none"> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure sensor failure</li> <li>Damaged catalytic converter</li> <li>Blocked exhaust system</li> </ul>                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> <li>Check throttle blade for carbon obstruction</li> <li>Check and install new manifold absolute pressure sensor as required</li> <li>Check catalytic converter for damage</li> <li>Check exhaust system for blockage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P012B-84 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit Range /Performance - Signal below allowable range | <ul style="list-style-type: none"> <li>Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure sensor failure</li> <li>Damaged catalytic converter</li> <li>Blocked exhaust system</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> <li>Check throttle blade for carbon obstruction</li> </ul>                                                                                                                 |

|          |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>▪ Check and install new manifold absolute pressure sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Check exhaust system for blockage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| P012B-85 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit Range /Performance - Signal above allowable range | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure sensor failure</li> <li>▪ Damaged catalytic converter</li> <li>▪ Blocked exhaust system</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Check exhaust system for blockage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P012C-00 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit Low - No sub                                      | <ul style="list-style-type: none"> <li>▪ Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|          |                                                                                         |                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | type information                                                                        | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector terminal is backed out, connector terminal corrosion</li> <li>■ Manifold absolute pressure sensor failure</li> </ul>                                                                                                                                          | <p>circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Check and install new manifold absolute pressure sensor as required</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                     |
| P012D-00 | Turbocharger /Supercharger Inlet Pressure Sensor Circuit High - No sub type information | <ul style="list-style-type: none"> <li>■ Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector terminal is backed out, connector terminal corrosion</li> <li>■ Manifold absolute pressure sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Check and install new manifold absolute pressure sensor as required</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0130-00 | O2 Sensor Circuit (Bank 1 Sensor 1) - No sub type information                           | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul>                                          | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul>                           |
| P0131-00 | O2 Sensor Circuit Low                                                                   |                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                                    |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Voltage (Bank 1 Sensor 1) -<br>No sub type information                             | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <p>heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul>                                                        |
| P0132-00 | O2 Sensor Circuit High Voltage (Bank 1 Sensor 1) -<br>No sub type information      | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul> |
| P0132-85 | O2 Sensor Circuit High Voltage (Bank 1 Sensor 1) -<br>Signal above allowable range | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul> |

|          |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0133-00 | O2 Sensor Circuit Slow Response (Bank 1 Sensor 1) - No sub type information        | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor degraded</li> <li>■ Heated oxygen sensor contamination by incorrect fuel or oil</li> <li>■ Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>■ Check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first</li> <li>■ Check heated oxygen sensor for age degradation</li> <li>■ Check heated oxygen sensor for contamination by incorrect fuel or oil</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul>                                                                               |
| P0134-00 | O2 Sensor Circuit No Activity Detected (Bank 1 Sensor 1) - No sub type information | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor incorrectly installed or missing</li> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul>                                                                                                                                                                             | <ul style="list-style-type: none"> <li>■ Check heated oxygen sensor is correctly installed</li> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul> |
| P0135-00 | O2 Sensor Heater Circuit (Bank 1 Sensor 1) - No sub type information               | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The heated oxygen sensor heater duty cycle is at its maximum value but the sensor does not reach operating temperature</p> </div> <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                |

|          |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                           | <ul style="list-style-type: none"> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P0136-00 | O2 Sensor Circuit (Bank 1 Sensor 2) - No sub type information             | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                    |
| P0137-00 | O2 Sensor Circuit Low Voltage (Bank 1 Sensor 2) - No sub type information | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Heated oxygen sensor voltage below the lower limit         </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy             <ul style="list-style-type: none"> <li>Fully warmed-up engine, drive off-idle for a minimum of 5 minutes</li> </ul> </li> <li>Prioritised Checks to Perform</li> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P0137-84 | O2 Sensor Circuit Low Voltage (Bank 1 Sensor 2) -                         | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|          |                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Signal below allowable range                                                    | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                     | <p>ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                               |
| P0138-00 | O2 Sensor Circuit High Voltage (Bank 1 Sensor 2) - No sub type information      | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Heated oxygen sensor voltage above the upper limit</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>Fully warmed-up engine, drive off-idle for a minimum of 5 minutes</li> </ul> </li> <li>Prioritised Checks to Perform</li> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P0138-85 | O2 Sensor Circuit High Voltage (Bank 1 Sensor 2) - Signal above allowable range | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic</li> </ul>                                                                                                                                                                                                                                                                                                                                                                  |

|          |                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| P013A-00 | O2 Sensor Slow Response - Rich to Lean (Bank 1 and Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>Exhaust system leakage</li> <li>Incorrectly installed heated oxygen sensor</li> <li>Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                      | <ul style="list-style-type: none"> <li>Check exhaust system for leakage</li> <li>Check heated oxygen sensors are installed correctly into the exhaust system</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Oxygen Sensor (O2S) Voltage Bank 1 Sensor 2 - (0x035F)</li> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P013B-00 | O2 Sensor Slow Response - Lean to Rich (Bank 1, Sensor 2) - No sub type information    | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Air leakage between heated oxygen sensor and catalyst</li> <li>Heated oxygen sensor tip damaged</li> <li>Heated oxygen sensor tip blocked</li> <li>Heated oxygen sensor tip contaminated by excessive oil consumption</li> <li>Heated oxygen sensor tip contaminated by poor or incorrect fuel</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal <ul style="list-style-type: none"> <li>Heated oxygen sensor signal for correct operation</li> </ul> </li> <li>Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check heated oxygen sensor is correctly installed</li> <li>Check heated oxygen sensor for tip damage</li> </ul>                                                                                                                                                                                                                                                                              |

|          |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>▪ Check heated oxygen sensor for tip blocked</li> <li>▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check</li> <li>▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| P013C-00 | O2 Sensor Slow Response - Rich to Lean (Bank 2, Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>▪ Exhaust system leakage</li> <li>▪ Incorrectly installed heated oxygen sensor</li> <li>▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check exhaust system for leakage</li> <li>▪ Check heated oxygen sensors are installed correctly into the exhaust system</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Oxygen Sensor (O2S) Voltage Bank 2 Sensor 2 - (0x0362)</li> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new heated oxygen sensor as required</li> </ul> |
| P013D-00 | O2 Sensor Slow Response - Lean to Rich (Bank 2, Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Air leakage between heated oxygen sensor and catalyst</li> <li>▪ Heated oxygen sensor tip damaged</li> <li>▪ Heated oxygen sensor tip blocked</li> </ul>                                                                   | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signal - Oxygen Sensor (O2S) Voltage Bank 2 Sensor 2 - (0x0362) <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor signal for correct operation</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|          |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                       | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor tip contaminated by excessive oil consumption</li> <li>▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel</li> </ul>                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Check heated oxygen sensor is correctly installed</li> <li>▪ Check heated oxygen sensor for tip damage</li> <li>▪ Check heated oxygen sensor for tip blocked</li> <li>▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check</li> <li>▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel</li> </ul>                                                                                                                                                                 |
| P013E-00 | O2 Sensor Delayed Response - Rich to Lean (Bank 1 Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>▪ Exhaust system leakage</li> <li>▪ Incorrectly installed heated oxygen sensor</li> <li>▪ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check exhaust system for leakage</li> <li>▪ Check heated oxygen sensors are installed correctly into the exhaust system</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Oxygen Sensor (O2S) Voltage Bank 1 Sensor 2 - (0x035F)</li> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new heated oxygen sensor as required</li> </ul> |

|          |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P013F-00 | O2 Sensor Delayed Response - Lean to Rich (Bank 1 Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Air leakage between heated oxygen sensor and catalyst</li> <li>■ Heated oxygen sensor tip damaged</li> <li>■ Heated oxygen sensor tip blocked</li> <li>■ Heated oxygen sensor tip contaminated by excessive oil consumption</li> <li>■ Heated oxygen sensor tip contaminated by poor or incorrect fuel</li> </ul> | <ul style="list-style-type: none"> <li>■ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Oxygen Sensor (O2S) Voltage Bank 1 Sensor 2 - (0x035F) <ul style="list-style-type: none"> <li>■ Heated oxygen sensor signal for correct operation</li> </ul> </li> <li>■ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Check heated oxygen sensor is correctly installed</li> <li>■ Check heated oxygen sensor for tip damage</li> <li>■ Check heated oxygen sensor for tip blocked</li> <li>■ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check</li> <li>■ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel</li> </ul> |
| P0141-00 | O2 Sensor Heater Circuit (Bank 1 Sensor 2) - No sub type information                  | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                |
| P0142-00 | O2 Sensor Circuit (Bank                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|          |                                                                                     |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | 1 Sensor 3) -<br>No sub type<br>information                                         | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <p>heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul>                                                        |
| P0143-00 | O2 Sensor<br>Circuit Low Voltage (Bank 1 Sensor 3) -<br>No sub type<br>information  | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul> |
| P0144-00 | O2 Sensor<br>Circuit High Voltage (Bank 1 Sensor 3) -<br>No sub type<br>information | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul> |

|          |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0147-00 | O2 Sensor Heater Circuit (Bank 1 Sensor 3) - No sub type information                  | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul>                                                                                          | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                  |
| P014A-00 | O2 Sensor Delayed Response - Rich to Lean (Bank 2 Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>■ Exhaust system leakage</li> <li>■ Incorrectly installed heated oxygen sensor</li> <li>■ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Check exhaust system for leakage</li> <li>■ Check heated oxygen sensors are installed correctly into the exhaust system</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Oxygen Sensor (O2S) Voltage Bank 2 Sensor 2 - (0x0362)</li> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul> |
| P014B-00 | O2 Sensor Delayed Response - Lean to Rich (Bank 2 Sensor 2) -                         | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Air leakage between heated oxygen sensor and catalyst</li> </ul>                                                                                                                                                           | <ul style="list-style-type: none"> <li>■ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals -</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|          |                                                                           |                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | No sub type information                                                   | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor tip damaged</li> <li>▪ Heated oxygen sensor tip blocked</li> <li>▪ Heated oxygen sensor tip contaminated by excessive oil consumption</li> <li>▪ Heated oxygen sensor tip contaminated by poor or incorrect fuel</li> </ul>  | <p>Oxygen Sensor (O2S)<br/>Voltage Bank 2 Sensor 2 - (0x0362)</p> <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor signal for correct operation</li> <li>▪ Refer to electrical circuit diagrams and check the heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Check heated oxygen sensor is correctly installed</li> <li>▪ Check heated oxygen sensor for tip damage</li> <li>▪ Check heated oxygen sensor for tip blocked</li> <li>▪ Check heated oxygen sensor for tip contamination by excessive oil consumption. Carry out oil consumption check</li> <li>▪ Check heated oxygen sensor for tip contamination by poor or incorrect fuel. Check customer is using correct grade of fuel</li> </ul> |
| P0150-00 | O2 Sensor Circuit (Bank 2 Sensor 1) - No sub type information             | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Heated oxygen sensor</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                 |
| P0151-00 | O2 Sensor Circuit Low Voltage (Bank 2 Sensor 1) - No sub type information | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                 | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|          |                                                                                    |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                    | <ul style="list-style-type: none"> <li>Heated oxygen sensor</li> </ul>                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                             |
| P0152-00 | O2 Sensor Circuit High Voltage (Bank 2 Sensor 1) - No sub type information         | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P0152-17 | O2 Sensor Circuit High Voltage (Bank 2 Sensor 1) - Circuit voltage above threshold | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P0153-00 | O2 Sensor Circuit Slow Response (Bank 2 Sensor 1) -                                | <ul style="list-style-type: none"> <li>Heated oxygen sensor degraded</li> <li>Heated oxygen sensor contamination by incorrect fuel or oil</li> </ul>                                                                                                                                 | <ul style="list-style-type: none"> <li>Check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first</li> </ul>                                                                                                                                                                                                                                                                                                                                                              |

|          |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | No sub type information                                                            | <ul style="list-style-type: none"> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Check heated oxygen sensor for age degradation</li> <li>Check heated oxygen sensor for contamination by incorrect fuel or oil</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                        |
| P0154-00 | O2 Sensor Circuit No Activity Detected (Bank 2 Sensor 1) - No sub type information | <ul style="list-style-type: none"> <li>Heated oxygen sensor incorrectly installed or missing</li> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Check heated oxygen sensor is correctly installed</li> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul>           |
| P0155-00 | O2 Sensor Heater Circuit (Bank 2 Sensor 1) - No sub type information               | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The heated oxygen sensor heater duty cycle is at its maximum value but the sensor does not reach operating temperature</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>Start engine and allow to idle</li> </ul> </li> <li>Prioritised Checks to Perform</li> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored</li> </ul> |

|          |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |
| P0155-85 | O2 Sensor Heater Circuit (Bank 2 Sensor 1) - Signal above allowable range | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The heated oxygen sensor heater duty cycle is at its maximum value but the sensor does not reach operating temperature</p> </div> <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P0156-00 | O2 Sensor Circuit (Bank 2 Sensor 2 - No sub type information)             | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P0157-00 | O2 Sensor Circuit Low Voltage (Bank 2 Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                    |

|          |                                                                            |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                            |                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                              |
| P0158-00 | O2 Sensor Circuit High Voltage (Bank 2 Sensor 2) - No sub type information | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul> |
| P0161-00 | O2 Sensor Heater Circuit (Bank 2 Sensor 2) - No sub type information       | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new heated oxygen sensor as required</li> </ul> |
| P0162-00 | O2 Sensor Circuit (Bank 2 Sensor 3) - No sub type information              | <ul style="list-style-type: none"> <li>■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and</li> </ul>                                                                                                                                                                                                                                         |

|          |                                                                            |                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                            |                                                                                                                                                                                                                                                                                              | <p>pins for damage and/or corrosion</p> <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                  |
| P0163-00 | O2 Sensor Circuit Low Voltage (Bank 2 Sensor 3) - No sub type information  | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P0164-00 | O2 Sensor Circuit High Voltage (Bank 2 Sensor 3) - No sub type information | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Heated oxygen sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new heated oxygen sensor as required</li> </ul> |
| P0167-00 | O2 Sensor Heater Circuit (Bank 2 Sensor 3) - No sub type information       | <ul style="list-style-type: none"> <li>Heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                       | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                 |

|          |                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                               | <ul style="list-style-type: none"> <li>▪ Heated oxygen sensor failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new heated oxygen sensor as required</li> </ul>                                                                                                                                                                                     |
| P0168-22 | Engine Fuel Temperature Too High - Signal amplitude > maximum | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The fuel temperature is compared against a modelled value. The DTC is set when the fuel temperature is greater than the modelled value</p> </div> <ul style="list-style-type: none"> <li>▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel temperature sensor as required</li> </ul> |
| P0169-00 | Incorrect Fuel Composition - No sub type information          | <ul style="list-style-type: none"> <li>▪ Desired fuelling control limits implausible. Engine control module software error</li> <li>▪ Engine control module failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module</li> <li>▪ Check and install new engine control module as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> </ul>          |

|          |                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0169-42 | Incorrect Fuel Composition - General memory failure | <ul style="list-style-type: none"> <li>Desired lambda was not found plausible. Engine control module software error</li> <li>Engine control module failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the power and ground connections to the module</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module</li> <li>Check and install new engine control module as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> </ul> |
| P0169-43 | Incorrect Fuel Composition - Special memory failure | <ul style="list-style-type: none"> <li>Predicted fuel mass was found not plausible. Engine control module software error</li> <li>Engine control module failure</li> </ul>                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the power and ground connections to the module</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module</li> <li>Check and install new engine control module as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> </ul> |
| P0170-00 | Fuel Trim (Bank 1) - No sub type information        | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Detects when secondary fuelling adaptation control signal is not plausible</p> </div> <ul style="list-style-type: none"> <li>Exhaust system leakage</li> <li>Post catalyst heated oxygen sensor incorrectly installed</li> <li>Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul> | <div>  <b>NOTE:</b> </div> <div> <p>Operational requirements needed to allow the monitor to be fully tested. Drive vehicle under steady state conditions for up to 20 minutes</p> </div> <ul style="list-style-type: none"> <li>Check exhaust system for leakage. Rectify as required</li> </ul>                                                                                                                             |

|          |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                     | <ul style="list-style-type: none"> <li>■ Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Pre catalyst heated oxygen sensor contamination or failure</li> <li>■ Post catalyst heated oxygen sensor contamination or failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>■ Check post catalyst heated oxygen sensor is correctly installed</li> <li>■ Refer to the electrical circuit diagrams and check pre catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |
| P0170-29 | Fuel Trim (Bank 1) - Signal invalid | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Detects when secondary fuelling adaptation offset is too large to maintain emissions within limits</p> </div> <ul style="list-style-type: none"> <li>■ Exhaust system leakage</li> <li>■ Post catalyst heated oxygen sensor incorrectly installed</li> <li>■ Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Pre catalyst heated oxygen sensor contamination or failure</li> <li>■ Post catalyst heated oxygen sensor contamination or failure</li> </ul> | <div>  <b>NOTE:</b> </div> <div> <p>Operational requirements needed to allow the monitor to be fully tested. Drive vehicle under steady state conditions for up to 20 minutes</p> </div> <ul style="list-style-type: none"> <li>■ Check exhaust system for leakage. Rectify as required</li> <li>■ Check post catalyst heated oxygen sensor is correctly installed</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check pre catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Refer to the electrical circuit diagrams and check post catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul> |

|          |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new pre catalyst heated oxygen sensor as required</li> <li>Check and install new post catalyst heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| P0170-64 | Fuel Trim (Bank 1) - Signal plausibility failure | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Detects when secondary fuelling adaptation offset is too large to maintain emissions within limits</p> </div> <ul style="list-style-type: none"> <li>Exhaust system leakage</li> <li>Post catalyst heated oxygen sensor incorrectly installed</li> <li>Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Pre catalyst heated oxygen sensor contamination or failure</li> <li>Post catalyst heated oxygen sensor contamination or failure</li> </ul> | <div>  <b>NOTE:</b> </div> <div> <p>Operational requirements needed to allow the monitor to be fully tested. Drive vehicle under steady state conditions for up to 20 minutes</p> </div> <ul style="list-style-type: none"> <li>Check exhaust system for leakage. Rectify as required</li> <li>Check post catalyst heated oxygen sensor is correctly installed</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check pre catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Refer to the electrical circuit diagrams and check post catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new pre catalyst heated oxygen sensor as required</li> </ul> |

|          |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Check and install new post catalyst heated oxygen sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| P0173-00 | Fuel Trim (Bank 2) - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Detects when secondary fuelling adaptation control signal is not plausible</p> </div> <ul style="list-style-type: none"> <li>Exhaust system leakage</li> <li>Post catalyst heated oxygen sensor incorrectly installed</li> <li>Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Pre catalyst heated oxygen sensor contamination or failure</li> <li>Post catalyst heated oxygen sensor contamination or failure</li> </ul> | <div>  <b>NOTE:</b> </div> <div> <p>Operational requirements needed to allow the monitor to be fully tested. Drive vehicle under steady state conditions for up to 20 minutes</p> </div> <ul style="list-style-type: none"> <li>Check exhaust system for leakage. Rectify as required</li> <li>Check post catalyst heated oxygen sensor is correctly installed</li> <li>Refer to the electrical circuit diagrams and check Heated Oxygen Sensor (HO2S) heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to Electronic Engine Controls V8 S/C 5.0L - Petrol - Heated Oxygen Sensors (HO2S), Diagnosis and Testing - pinpoint test A</li> </ul> |
| P0173-29 | Fuel Trim (Bank 2) - Signal invalid          | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Detects when secondary fuelling adaptation offset is too large to maintain emissions within limits</p> </div> <ul style="list-style-type: none"> <li>Exhaust system leakage</li> <li>Post catalyst heated oxygen sensor incorrectly installed</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                     | <div>  <b>NOTE:</b> </div> <div> <p>Operational requirements needed to allow the monitor to be fully tested. Drive vehicle under steady state conditions for up to 20 minutes</p> </div> <ul style="list-style-type: none"> <li>Check exhaust system for leakage. Rectify as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|          |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                  | <ul style="list-style-type: none"> <li>■ Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Pre catalyst heated oxygen sensor contamination or failure</li> <li>■ Post catalyst heated oxygen sensor contamination or failure</li> </ul>                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>■ Check post catalyst heated oxygen sensor is correctly installed</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check pre catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Refer to the electrical circuit diagrams and check post catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new pre catalyst heated oxygen sensor as required</li> <li>■ Check and install new post catalyst heated oxygen sensor as required</li> </ul> |
| P0173-64 | Fuel Trim (Bank 2) - Signal plausibility failure | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Detects when secondary fuelling adaptation offset is too small to maintain emissions within limits</p> </div> <ul style="list-style-type: none"> <li>■ Exhaust system leakage</li> <li>■ Post catalyst heated oxygen sensor incorrectly installed</li> <li>■ Pre catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Post catalyst heated oxygen sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Pre catalyst heated oxygen sensor contamination or failure</li> </ul> | <div>  <b>NOTE:</b> </div> <div> <p>Operational requirements needed to allow the monitor to be fully tested. Drive vehicle under steady state conditions for up to 20 minutes</p> </div> <ul style="list-style-type: none"> <li>■ Check exhaust system for leakage. Rectify as required</li> <li>■ Check post catalyst heated oxygen sensor is correctly installed</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check pre catalyst heated oxygen</li> </ul>                                                                                                                                                                                                                                                    |

|          |                                                                                 |                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                 | <ul style="list-style-type: none"> <li>Post catalyst heated oxygen sensor contamination or failure</li> </ul>                                                                                                                                                                                                                                                         | <p>sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check post catalyst heated oxygen sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new pre catalyst heated oxygen sensor as required</li> <li>Check and install new post catalyst heated oxygen sensor as required</li> </ul> |
| P0174-00 | System Too Lean (Bank 2)<br>- No sub type information                           | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul>                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                          |
| P0177-34 | Flexible Fuel Sensor Circuit Range /Performance<br>- Signal high time < minimum | <ul style="list-style-type: none"> <li>Electromagnetic compatibility interference</li> <li>Other related DTCs</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Flex fuel sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Check for additional after market installations that may cause electromagnetic compatibility interference</li> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                            |

|          |                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>▪ Check and install new flex fuel sensor as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| P0177-35 | Flexible Fuel Sensor Circuit Range /Performance<br>- Signal high time > maximum | <ul style="list-style-type: none"> <li>▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value</li> <li>▪ Electromagnetic compatibility interference</li> <li>▪ Other related DTCs</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Flex fuel sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check for additional after market installations that may cause electromagnetic compatibility interference</li> <li>▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Check and install new flex fuel sensor as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0177-36 | Flexible Fuel Sensor Circuit Range /Performance<br>- Signal frequency too low   | <ul style="list-style-type: none"> <li>▪ The engine control module detected excessive duration for one cycle of the output across a specified sample size</li> <li>▪ Electromagnetic compatibility interference</li> <li>▪ Other related DTCs</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Flex fuel sensor failure</li> </ul>   | <ul style="list-style-type: none"> <li>▪ Check for additional after market installations that may cause electromagnetic compatibility interference</li> <li>▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Check and install new flex fuel sensor as required</li> </ul>                                                                                                                                      |

|          |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| P0177-37 | Flexible Fuel Sensor Circuit Range /Performance<br>- Signal frequency too high | <ul style="list-style-type: none"> <li>The engine control module detected insufficient duration for one cycle of the output across a specified sample size</li> <li>Electromagnetic compatibility interference</li> <li>Other related DTCs</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Flex fuel sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Check for additional after market installations that may cause electromagnetic compatibility interference</li> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install new flex fuel sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0177-62 | Flexible Fuel Sensor Circuit Range /Performance<br>- Signal compare failure    | <ul style="list-style-type: none"> <li>The engine control module detected failure when comparing two or more input parameters for plausibility</li> <li>Other related DTCs</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Flex fuel sensor failure</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install new flex fuel sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                    |
|          |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0177-64 | Flexible Fuel Sensor Circuit Range /Performance - Signal plausibility failure | <ul style="list-style-type: none"> <li>Other related DTCs</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Flex fuel sensor failure</li> </ul>                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install new flex fuel sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0178-11 | Flexible Fuel Sensor Circuit Low - Circuit short to ground                    | <ul style="list-style-type: none"> <li>The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected</li> <li>Other related DTCs</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Flex fuel sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Check and install new flex fuel sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0179-00 | Flexible Fuel Sensor Circuit High - No sub type information                   | <ul style="list-style-type: none"> <li>Other related DTCs</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Flex fuel sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Flex fuel sensor failure</li> </ul>                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to electrical circuit diagrams and check the flex fuel sensor circuit for short circuit to ground,</li> </ul>                                                                                                                                                                                                                                                      |

|          |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>▪ Check and install new flex fuel sensor as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                  |
| P0181-21 | Fuel Temperature Sensor A<br>Circuit Range /Performance<br>- Signal amplitude < minimum | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Electrical physical range check of fuel temperature sensor</p> </div> <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel temperature sensor failure</li> </ul>   | <ul style="list-style-type: none"> <li>▪ Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>▪ Ignition on</li> </ul> </li> <li>▪ Prioritised Checks to Perform</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel temperature sensor as required</li> </ul> |
| P0181-22 | Fuel Temperature Sensor A<br>Circuit Range /Performance<br>- Signal amplitude > maximum | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Electrical physical range check of fuel temperature sensor</p> </div> <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>▪ Ignition on</li> </ul> </li> <li>▪ Prioritised Checks to Perform</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored</li> </ul>                                                                                                                     |

|          |                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>Check and install new fuel temperature sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| P0181-84 | Fuel Temperature Sensor A Circuit Range /Performance<br>- Signal below allowable range | <div>  <b>NOTE:</b> </div> <div> <p>Comparison of fuel temperature against modelled is less than modelled value</p> </div> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel temperature sensor failure <ul style="list-style-type: none"> <li>Possible contamination of sensor or connector keeping sensor stuck low, series resistance</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>Drive vehicle under steady-state conditions for up to 30 minutes. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Fuel Temperature "A" - Raw - (0x0429)</li> </ul> </li> <li>Prioritised Checks to Perform</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel temperature sensor as required</li> </ul> |
| P0181-85 | Fuel Temperature Sensor A Circuit Range /Performance<br>- Signal above allowable range | <div>  <b>NOTE:</b> </div> <div> <p>Comparison of fuel temperature against modelled is greater than modelled value</p> </div> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel temperature sensor failure</li> </ul>                                                                                                                                         | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>Drive vehicle under steady-state conditions for up to 30 minutes. Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Fuel Temperature "A" - Raw - (0x0429)</li> </ul> </li> <li>Prioritised Checks to Perform</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|          |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                  | <ul style="list-style-type: none"> <li>▪ Possible contamination of sensor or connector keeping sensor stuck high, parallel resistance</li> </ul>                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel temperature sensor as required</li> </ul>                                                                                                          |
| P0182-00 | Fuel Temperature Sensor A Circuit Low - No sub type information  | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel rail pressure sensor 5 volt power supply circuit, open circuit, high resistance</li> <li>▪ Fuel temperature sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Check fuel rail pressure sensor 5 volt power supply circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel temperature sensor as required</li> </ul> |
| P0183-00 | Fuel Temperature Sensor A Circuit High - No sub type information | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel temperature sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel temperature sensor failure</li> </ul>                                                                                                 | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs and refer to this DTC index. Rectify these first</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                           |

|          |                                                                                      |                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                      |                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel temperature sensor as required</li> </ul>                                                                                                                                                                                    |
| P0191-21 | Fuel Rail Pressure Sensor A Circuit Range /Performance - Signal amplitude < minimum  | <ul style="list-style-type: none"> <li>Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel rail pressure sensor failure</li> </ul>                                                                                                                          | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel rail pressure sensor as required</li> </ul> |
| P0191-64 | Fuel Rail Pressure Sensor A Circuit Range /Performance - Signal plausibility failure | <ul style="list-style-type: none"> <li>Fuel rail pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel rail pressure sensor failure</li> </ul>                                                                                                                          | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel rail pressure sensor as required</li> </ul> |
| P0191-71 | Fuel Rail Pressure Sensor A Circuit Range /Performance - Actuator stuck              | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> The fuel rail pressure is sampled over 3 injections, if no difference is seen then the DTC is set</p> </div> <ul style="list-style-type: none"> <li>Fuel rail pressure sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Check and install new fuel rail pressure sensor as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                  |
| P0192-00 | Fuel Rail Pressure Sensor A                                                          | <div>  <b>NOTE:</b> </div>                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy</li> </ul>                                                                                                                                                                                                                                                                                                                               |

|          |                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Circuit Low -<br>No sub type<br>information                               | <p><b>Monitor description.</b> High pressure fuel pressure sensor voltage below threshold</p> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel rail pressure sensor circuit, short circuit to ground</li> <li>Fuel rail pressure sensor 5 volt power supply circuit, open circuit, high resistance</li> <li>Fuel rail pressure sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Ignition on, start engine and idle for 30 seconds</li> <li>Prioritised Checks to Perform</li> <li>Check connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground</li> <li>Check fuel rail pressure sensor 5 volt power supply circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel rail pressure sensor as required</li> </ul> |
| P0192-84 | Fuel Rail Pressure Sensor A<br>Circuit Low - Signal below allowable range | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel rail pressure sensor circuit, short circuit to ground</li> <li>Fuel rail pressure sensor 5 volt power supply circuit, open circuit, high resistance</li> <li>Fuel rail pressure sensor failure</li> </ul>                                                                                               | <ul style="list-style-type: none"> <li>Check connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground</li> <li>Check fuel rail pressure sensor 5 volt power supply circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel rail pressure sensor as required</li> </ul>                                                                                                   |
| P0193-   | Fuel Rail                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|          |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00       | Pressure Sensor A Circuit High - No sub type information                | <p><b>NOTE:</b></p> <p><b>Monitor description.</b> High pressure fuel pressure sensor voltage above threshold</p> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel rail pressure sensor circuit, short circuit to power</li> <li>Fuel rail pressure sensor ground supply circuit, open circuit, high resistance</li> <li>Fuel rail pressure sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>Ignition on, start engine and idle for 30 seconds</li> </ul> </li> <li>Prioritised Checks to Perform</li> <li>Check connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power</li> <li>Check fuel rail pressure sensor ground supply circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel rail pressure sensor as required</li> </ul> |
| P0193-85 | Fuel Rail Pressure Sensor A Circuit High - Signal above allowable range | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel rail pressure sensor circuit, short circuit to power</li> <li>Fuel rail pressure sensor ground supply circuit, open circuit, high resistance</li> <li>Fuel rail pressure sensor failure</li> </ul>                                                                                                                   | <ul style="list-style-type: none"> <li>Check connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power</li> <li>Check fuel rail pressure sensor ground supply circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel rail pressure sensor as required</li> </ul>                                                                                                                                                                                                        |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0194-00 | Fuel Rail Pressure Sensor Circuit Intermittent /Erratic Bank 1 - No sub type information | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel rail pressure sensor circuit, short circuit to power</li> <li>Fuel rail pressure sensor ground supply circuit, open circuit, high resistance</li> <li>Fuel rail pressure sensor failure</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>Check connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power</li> <li>Check fuel rail pressure sensor ground supply circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>Check and install a new fuel rail pressure sensor as required</li> </ul> |
| P0195-00 | Engine Oil Temperature Sensor Circuit - No sub type information                          | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Engine control module monitors the oil temperature from the sensor and sets the DTC if the temperature is not plausible</p> </div> <ul style="list-style-type: none"> <li>Oil level sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Oil level sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check oil level sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new oil level sensor as required</li> </ul>                                                                                                       |
| P0201-13 | Cylinder 1 Injector Circuit / Open - Circuit open                                        | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                                           | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                       |

|          |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                   | <ul style="list-style-type: none"> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |
| P0202-13 | Cylinder 2 Injector Circuit / Open - Circuit open | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>   | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0203-13 | Cylinder 3 Injector Circuit / Open - Circuit open | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0204-13 | Cylinder 4 Injector Circuit / Open - Circuit open | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                                                                                  | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic</li> </ul>                                                                                                                                                                                |

|          |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                   | <ul style="list-style-type: none"> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                 | <p>equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                          |
| P0205-13 | Cylinder 5 Injector Circuit / Open - Circuit open | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>   | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0206-13 | Cylinder 6 Injector Circuit / Open - Circuit open | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0207-13 | Cylinder 7 Injector Circuit / Open - Circuit open | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div>                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                           |

|          |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                               | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, open circuit, high resistance</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                  |
| P0208-13 | Cylinder 8 Injector Circuit / Open - Circuit open                                             | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, open circuit, high resistance</li> <li>Fuel injector failure</li> </ul>                                                                                                              | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul> |
| P0219-00 | Engine Overspeed Condition - No sub type information                                          | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The DTC is set if the engine speed is greater than 7100rpm for greater than 1 second</p> </div> <ul style="list-style-type: none"> <li>Incorrect gear was selected</li> <li>Automatic transmission failure</li> </ul>                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Check engine control module for related DTCs and refer to this DTC index. Rectify these first</li> <li>Check transmission control module for related DTCs and refer to the relevant DTC index</li> </ul>                                                                                                                                                                                                                                               |
| P0221-00 | Throttle/Pedal Position Sensor /Switch B Circuit Range /Performance - No sub type information | <ul style="list-style-type: none"> <li>Electrical Cause <ul style="list-style-type: none"> <li>Yes</li> </ul> </li> <li>Mechanical Cause <ul style="list-style-type: none"> <li>No</li> </ul> </li> <li>Control module cavity <ul style="list-style-type: none"> <li>Circuit reference I_A_TVA2</li> </ul> </li> <li>Monitor Description <ul style="list-style-type: none"> <li>Throttle position sensor 2 does not align to position sensor 1 and is also not aligned to a theoretical throttle angle calculated from air charge / load. Throttle position sensor 2 is electrically within valid range</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging Strategy <ul style="list-style-type: none"> <li>Ensure all throttle position circuit fault have been rectified prior to investigating this fault</li> </ul> </li> <li>Engine running at idle for 5 seconds, also perform a short (less than 10 minute) varied drive cycle to ensure fault has been rectified, ensure light</li> </ul>                                                                         |

|          |                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                      | <p>but irrational to operating conditions</p> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Electric throttle - Wiring integrity <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>and high engine loads are included</p> <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment, with ignition on but engine off, check electric throttle position sensor signal 2 is aligned to electric throttle position sensor signal 1</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check the electric throttle circuit for <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0222-00 | Throttle/Pedal Position Sensor /Switch B<br>Circuit Low -<br>No sub type information | <ul style="list-style-type: none"> <li>Electrical Cause <ul style="list-style-type: none"> <li>Yes</li> </ul> </li> <li>Mechanical Cause <ul style="list-style-type: none"> <li>No</li> </ul> </li> <li>Control module cavity <ul style="list-style-type: none"> <li>Circuit reference I_A_TVA2 (engine control module)</li> <li>Circuit reference POT 2 (electric throttle)</li> </ul> </li> <li>Monitor Description <ul style="list-style-type: none"> <li>Throttle position sensor 2 is below valid electrical range</li> </ul> </li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Electric throttle - Wiring integrity <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging Strategy <ul style="list-style-type: none"> <li>Ignition On, Engine greater than 1200 rpm for 5 seconds</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check the electric throttle circuit for <ul style="list-style-type: none"> <li>Short circuit to ground</li> <li>Short circuit to power</li> <li>Open circuit</li> <li>High resistance</li> </ul> </li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                    |
| P0223-85 | Throttle/Pedal Position Sensor /Switch B                                             | <ul style="list-style-type: none"> <li>Electrical Cause <ul style="list-style-type: none"> <li>Yes</li> </ul> </li> <li>Mechanical Cause</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging Strategy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|          |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Circuit High - Signal above allowable range                                | <ul style="list-style-type: none"> <li>▪ No</li> <li>▪ Control module cavity <ul style="list-style-type: none"> <li>▪ Circuit reference I_A_TVA2 (engine control module)</li> <li>▪ Circuit reference POT 2 (electric throttle)</li> </ul> </li> <li>▪ Monitor Description <ul style="list-style-type: none"> <li>▪ Throttle position sensor 2 is above valid electrical range</li> </ul> </li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Electric throttle - Wiring integrity <ul style="list-style-type: none"> <li>▪ Short circuit to ground</li> <li>▪ Short circuit to power</li> <li>▪ Open circuit</li> <li>▪ High resistance</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>▪ Ignition On, Engine greater than 1200 rpm for 5 seconds</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check the electric throttle circuit for <ul style="list-style-type: none"> <li>▪ Short circuit to ground</li> <li>▪ Short circuit to power</li> <li>▪ Open circuit</li> <li>▪ High resistance</li> </ul> </li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0230-13 | Fuel Pump Primary Circuit - Circuit open                                   | <ul style="list-style-type: none"> <li>▪ The diagnostic detects failures between the fuel pump driver module and the fuel pump but reports them to the powertrain control module via the PWM line</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Harness failure - Wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 |
| P0230-19 | Fuel Pump Primary Circuit - Circuit current above threshold                | <ul style="list-style-type: none"> <li>▪ The diagnostic detects failures between the fuel pump driver module and the fuel pump but reports them to the powertrain control module via the PWM line</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Harness failure - Wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 |
| P0234-00 | Turbocharger /Supercharger A Overboost Condition - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> If the pressure ratio across the supercharger and bypass is above the expected amount the DTC is set</p> </div> <ul style="list-style-type: none"> <li>▪ Intake manifold pressure sensors incorrect readings</li> </ul>                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check datalogger signals - Boost Absolute Pressure - Raw Value - (0x033E)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                           |

|          |                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0236-21 | Turbocharger /Supercharger Boost Sensor A Circuit Range /Performance - Signal amplitude < minimum | <ul style="list-style-type: none"> <li>Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure sensor failure</li> <li>Damaged catalytic converter</li> <li>Blocked exhaust system</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> <li>Check air filter for blockage or restriction</li> <li>Check engine breather system for leakage</li> <li>Check throttle blade for carbon obstruction</li> <li>Check and install new manifold absolute pressure sensor as required</li> <li>Check catalytic converter for damage</li> <li>Check exhaust system for blockage</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0236-22 | Turbocharger /Supercharger Boost Sensor A Circuit Range /Performance - Signal amplitude > maximum | <ul style="list-style-type: none"> <li>Manifold absolute pressure sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Intake manifold air leak</li> <li>Throttle adaption failure</li> <li>Blocked air filter</li> <li>Engine breather leakage</li> <li>Carbon obstruction around throttle blade</li> <li>Manifold absolute pressure sensor failure</li> <li>Damaged catalytic converter</li> <li>Blocked exhaust system</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>Check intake manifold for leakage, loose or missing components</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                          |                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Check exhaust system for blockage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0236-23 | Turbocharger /Supercharger Boost Sensor A Circuit Range /Performance - Signal stuck low  | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Harness failure - Wiring integrity</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                       |
| P0236-24 | Turbocharger /Supercharger Boost Sensor A Circuit Range /Performance - Signal stuck high | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Harness failure - Wiring integrity</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                       |
| P0236-29 | Turbocharger /Supercharger Boost Sensor A Circuit Range /Performance - Signal invalid    | <ul style="list-style-type: none"> <li>▪ The value of the signal measured by the engine control module is not plausible given the operating conditions</li> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                          |

|          |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                     | <ul style="list-style-type: none"> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul>                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure and temperature sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0236-84 | Turbocharger /Supercharger Boost Sensor A Circuit Range /Performance - Signal below allowable range | <ul style="list-style-type: none"> <li>▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range</li> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to ground</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure</li> </ul>                                                                     |

|          |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>and temperature sensor as required</p> <ul style="list-style-type: none"> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| P0236-85 | Turbocharger /Supercharger Boost Sensor A Circuit Range /Performance - Signal above allowable range | <ul style="list-style-type: none"> <li>▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range</li> <li>▪ Manifold absolute pressure and temperature sensor circuit, for short circuit to power</li> <li>▪ Intake manifold air leak</li> <li>▪ Throttle adaption failure</li> <li>▪ Blocked air filter</li> <li>▪ Engine breather leakage</li> <li>▪ Carbon obstruction around throttle blade</li> <li>▪ Manifold absolute pressure and temperature sensor failure</li> <li>▪ Damaged catalytic converter</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to power</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check throttle position voltage at ignition on</li> <li>▪ Check intake manifold for leakage, loose or missing components</li> <li>▪ Check air filter for blockage or restriction</li> <li>▪ Check engine breather system for leakage</li> <li>▪ Check throttle blade for carbon obstruction</li> <li>▪ Check and install new manifold absolute pressure and temperature sensor as required</li> <li>▪ Check catalytic converter for damage</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0237-00 | Turbocharger /Supercharger Boost Sensor A Circuit Low - No sub type information                     | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Harness failure - Wiring integrity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>▪ Using the Jaguar Land Rover approved diagnostic</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|          |                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| P0238-00 | Turbocharger /Supercharger Boost Sensor A Circuit High - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Circuit check of the manifold absolute pressure and temperature sensor</p> <ul style="list-style-type: none"> <li>Manifold absolute pressure and temperature sensor circuit, for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Harness failure - Wiring integrity</li> </ul> </div> | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>Ignition on</li> </ul> </li> <li>Prioritised Checks to Perform</li> <li>Refer to the electrical circuit diagrams and check manifold absolute pressure and temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P023A-13 | Charge Air Cooler Coolant Pump Control Circuit /Open - Circuit open              | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Air charge coolant pump relay circuit open circuit, high resistance</li> <li>Air charge coolant pump relay failure</li> </ul>                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check air charge coolant pump relay circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new air charge coolant pump relay as required</li> </ul>                                                                                                                                                                                                                                                                      |
| P023B-11 | Charge Air Cooler Coolant Pump Control Circuit Low - Circuit short to ground     | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Air charge coolant pump relay circuit short circuit to ground</li> <li>Air charge coolant pump relay failure</li> </ul>                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check air charge coolant pump</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|          |                                                                                     |                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                     |                                                                                                                                                                                                                                                                                                              | <p>relay circuit for short circuit to ground</p> <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new air charge coolant pump relay as required</li> </ul>                                                                                                                                                                          |
| P023C-12 | Charge Air Cooler Coolant Pump Control Circuit High - Circuit short to battery      | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Air charge coolant pump relay circuit short circuit to power</li> <li>Air charge coolant pump relay failure</li> </ul>                                                       | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check air charge coolant pump relay circuit for short circuit to power</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new air charge coolant pump relay as required</li> </ul> |
| P0251-13 | Injection Pump Fuel Metering Control A - Circuit open                               | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump circuit, open circuit</li> <li>Fuel pump failure</li> </ul>                                                                                                        | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump circuit for open circuit</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel pump as required</li> </ul>                                                   |
| P0252-11 | Injection Pump Fuel Metering Control A Range /Performance - Circuit short to ground | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> To diagnose an internal serial peripheral interface communication failure between the main central processing unit and the pump control module</p> </div> | <ul style="list-style-type: none"> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine control module as required</li> </ul>                                                                                                                     |

|          |                                                                        |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                        | <ul style="list-style-type: none"> <li>Engine control module failure</li> </ul>                                                                                                                                  | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> </ul>                                                                                                                                                                                                                                                       |
| P0253-11 | Injection Pump Fuel Metering Control A Low - Circuit short to ground   | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump circuit, short circuit to ground</li> <li>Fuel pump failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump circuit for short circuit to ground</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel pump as required</li> </ul> |
| P0254-12 | Injection Pump Fuel Metering Control A High - Circuit short to battery | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump circuit, short circuit to power</li> <li>Fuel pump failure</li> </ul>  | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump circuit for short circuit to power</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel pump as required</li> </ul>  |
| P0256-13 | Injection Pump Fuel Metering Control B - Circuit open                  | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump circuit, open circuit</li> <li>Fuel pump failure</li> </ul>            | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump circuit for open circuit</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel pump as required</li> </ul>            |
|          |                                                                        |                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|          |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0257-11 | Injection Pump Fuel Metering Control B Range /Performance - Circuit short to ground | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> To diagnose an internal serial peripheral interface communication failure between the main central processing unit and the pump control module</p> </div> <ul style="list-style-type: none"> <li>Engine control module failure</li> </ul> | <ul style="list-style-type: none"> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new engine control module as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> </ul> |
| P0258-11 | Injection Pump Fuel Metering Control B Low - Circuit short to ground                | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump circuit, short circuit to ground</li> <li>Fuel pump failure</li> </ul>                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump circuit for short circuit to ground</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel pump as required</li> </ul>                                                             |
| P0259-12 | Injection Pump Fuel Metering Control B High - Circuit short to battery              | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump circuit, short circuit to power</li> <li>Fuel pump failure</li> </ul>                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump circuit for short circuit to power</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel pump as required</li> </ul>                                                              |
| P025A-13 | Fuel Pump Module "A" Control Circuit /Open - Circuit open                           | <div>  <b>NOTE:</b> </div>                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check</li> </ul>                                                                                                                                                                                                                                                                                                   |

|          |                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                | <p><b>Monitor description.</b> To diagnose a failure on the signal line to the fuel pump driver module</p> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump driver module circuit, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                 | <p>fuel pump driver module circuit for open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump</li> </ul>                                                                                                                                                                                    |
| P025B-98 | Fuel Pump Module "A" Control Circuit Range /Performance - Component or system over temperature | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> To diagnose a failure on the PWM line from the powertrain control module to the fuel pump driver module</p> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump driver module circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel pump driver module failure</li> </ul> | <ul style="list-style-type: none"> <li>Check powertrain control module for related DTCs and refer to this DTC index. Rectify these first</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump driver module circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>Check and install a new fuel pump driver module as required</li> </ul> |
| P025C-11 | Fuel Pump Module "A" Control Circuit Low - Circuit short to ground                             | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> To diagnose a failure on the signal line to the fuel pump driver module</p> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump driver module circuit, short circuit to ground</li> </ul>                                                                                                                               | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump driver module circuit for short circuit to ground</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump</li> </ul>                                        |

|          |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P025D-12 | Fuel Pump Module "A" Control Circuit High - Circuit short to battery | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> To diagnose a failure on the signal line to the fuel pump driver module</p> </div> <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel pump driver module circuit, short circuit to power</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel pump driver module circuit for short circuit to power</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Inline diagnostic unit 2 non-intrusive test - Low pressure fuel pump</li> </ul> |
| P0261-11 | Cylinder 1 Injector Circuit Low - Circuit short to ground            | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, short circuit to ground</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>                                                                    |
| P0261-12 | Cylinder 1 Injector Circuit Low - Circuit short to battery           | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, short circuit to power</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>                                                                     |

|          |                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0261-13 | Cylinder 1<br>Injector "A"<br>Circuit Low -<br>Circuit open                 | <div>  <b>NOTES:</b> </div> <ul style="list-style-type: none"> <li>▪ <b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</li> <li>▪ This DTC is set when the powertrain control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output.</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul>                                                  |
| P0261-1C | Cylinder 1<br>Injector "A"<br>Circuit Low -<br>Circuit voltage out of range | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul> |
| P0262-11 | Cylinder 1<br>Injector<br>Circuit High -<br>Circuit short to ground         | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic</li> </ul>                                                                                                                                                                                                    |

|          |                                                                     |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                     |                                                                                                                                                                                                                                                                                      | <p>equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                        |
| P0262-12 | Cylinder 1 Injector Circuit High - Circuit short to battery         | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>                                                        | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                      |
| P0262-1C | Cylinder 1 Injector "A" Circuit High - Circuit voltage out of range | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul> |
| P0264-11 | Cylinder 2 Injector Circuit Low - Circuit short to ground           | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                       | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> </ul>                                                                                                                                                                                                                                                               |

|          |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                  |
| P0264-12 | Cylinder 2 Injector Circuit Low - Circuit short to battery         | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, short circuit to power</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>        |
| P0264-13 | Cylinder 2 Injector Circuit Low - Circuit open                     | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, open circuit, high resistance</li> <li>Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul> |
| P0264-1C | Cylinder 2 Injector "A" Circuit Low - Circuit voltage out of range | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                         |

|          |                                                             |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                             | <ul style="list-style-type: none"> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>                                      | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul>                                                                            |
| P0265-11 | Cylinder 2 Injector Circuit High - Circuit short to ground  | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0265-12 | Cylinder 2 Injector Circuit High - Circuit short to battery | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>  | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>  |
| P0265-1C | Cylinder 2 Injector "A" Circuit High - Circuit              | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                                                                            | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |

|          |                                                            |                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | voltage out of range                                       | <ul style="list-style-type: none"> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>                                      | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul>                                                                            |
| P0267-11 | Cylinder 3 Injector Circuit Low - Circuit short to ground  | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0267-12 | Cylinder 3 Injector Circuit Low - Circuit short to battery | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>  | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>  |
| P0267-13 | Cylinder 3 Injector Circuit Low - Circuit open             | <div>  <b>NOTE:</b> </div>                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |

|          |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                    | <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> <ul style="list-style-type: none"> <li>▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                              |
| P0267-1C | Cylinder 3 Injector "A" Circuit Low - Circuit voltage out of range | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul> |
| P0268-11 | Cylinder 3 Injector Circuit High - Circuit short to ground         | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                     |

|          |                                                                        |                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0268-12 | Cylinder 3 Injector Circuit High - Circuit short to battery            | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Fuel injector circuit, short circuit to power</li> <li>■ Fuel injector failure</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check fuel injector resistance</li> <li>■ Check and install new fuel injector as required</li> </ul>                      |
| P0268-1C | Cylinder 3 Injector "A" Circuit High - Circuit voltage out of range    | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Fuel injector failure</li> </ul>          | <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>■ Check fuel injector resistance</li> <li>■ Check and install a new fuel injector as required</li> </ul> |
| P026A-00 | Charge Air Cooler Efficiency Below Threshold - No sub type information | <ul style="list-style-type: none"> <li>■ Loss of coolant</li> <li>■ Air charge coolant pump relay</li> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Air charge coolant pump circuit</li> <li>■ Air charge coolant pump failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Check the engine coolant level is correct</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Check and install new air charge coolant pump relay as required</li> <li>■ Refer to the electrical circuit diagrams and check air charge coolant pump circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                              |

|          |                                                            |                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                            |                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new air charge coolant pump as required <ul style="list-style-type: none"> <li>Is the impeller damaged. Does the impeller rotate and pump coolant</li> </ul> </li> </ul>                                                                                                                    |
| P0270-11 | Cylinder 4 Injector Circuit Low - Circuit short to ground  | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, short circuit to ground</li> <li>Fuel injector failure</li> </ul>                                                      | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul> |
| P0270-12 | Cylinder 4 Injector Circuit Low - Circuit short to battery | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, short circuit to power</li> <li>Fuel injector failure</li> </ul>                                                       | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>  |
| P0270-13 | Cylinder 4 Injector Circuit Low - Circuit open             | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit </div> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check</li> </ul>                                                                                                                                                                                                                                                                                       |

|          |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                    | <div></div> <ul style="list-style-type: none"> <li>▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <p>fuel injector circuit for open circuit, high resistance</p> <ul style="list-style-type: none"> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                                                                                     |
| P0270-1C | Cylinder 4 Injector "A" Circuit Low - Circuit voltage out of range | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                         | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul> |
| P0271-11 | Cylinder 4 Injector Circuit High - Circuit short to ground         | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                     |
| P0271-12 | Cylinder 4 Injector Circuit High -                                 | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|          |                                                                     |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Circuit short to battery                                            | <ul style="list-style-type: none"> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                   | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                      |
| P0271-1C | Cylinder 4 Injector "A" Circuit High - Circuit voltage out of range | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul> |
| P0273-11 | Cylinder 5 Injector Circuit Low - Circuit short to ground           | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                       | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                     |

|          |                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0273-12 | Cylinder 5 Injector<br>Circuit Low -<br>Circuit short to battery         | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, short circuit to power</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>        |
| P0273-13 | Cylinder 5 Injector<br>Circuit Low -<br>Circuit open                     | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, open circuit, high resistance</li> <li>Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul> |
| P0273-1C | Cylinder 5 Injector "A"<br>Circuit Low -<br>Circuit voltage out of range | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> </ul>                                                                                      |

|          |                                                                     |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                     |                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                   |
| P0274-11 | Cylinder 5 Injector Circuit High - Circuit short to ground          | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                       | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0274-12 | Cylinder 5 Injector Circuit High - Circuit short to battery         | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>                                                        | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>  |
| P0274-1C | Cylinder 5 Injector "A" Circuit High - Circuit voltage out of range | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> </ul>                                                                                    |

|          |                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                   |
| P0276-11 | Cylinder 6 Injector Circuit Low - Circuit short to ground  | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0276-12 | Cylinder 6 Injector Circuit Low - Circuit short to battery | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>  |
| P0276-13 | Cylinder 6 Injector Circuit Low - Circuit open             | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                |

|          |                                                                    |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                    | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul>                                                 | <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                         |
| P0276-1C | Cylinder 6 Injector "A" Circuit Low - Circuit voltage out of range | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul> |
| P0277-11 | Cylinder 6 Injector Circuit High - Circuit short to ground         | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                       | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                     |
| P0277-12 | Cylinder 6 Injector Circuit High - Circuit short to battery        | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>                                                        | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic</li> </ul>                                                                                                                                                                                                     |

|          |                                                                     |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                     |                                                                                                                                                                                                                                                                                      | <p>equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                        |
| P0277-1C | Cylinder 6 Injector "A" Circuit High - Circuit voltage out of range | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install a new fuel injector as required</li> </ul> |
| P0279-11 | Cylinder 7 Injector Circuit Low - Circuit short to ground           | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                       | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                     |
| P0279-12 | Cylinder 7 Injector Circuit Low - Circuit short to battery          | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>                                                        | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> </ul>                                                                                                                                                                                                                                                                |

|          |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                                |
| P0279-13 | Cylinder 7 Injector "A" Circuit Low - Circuit open                 | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit, open circuit, high resistance</li> <li>Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check fuel injector resistance</li> <li>Check and install new fuel injector as required</li> </ul>               |
| P0279-1C | Cylinder 7 Injector "A" Circuit Low - Circuit voltage out of range | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>Check fuel injector resistance</li> <li>Check and install a new fuel injector as required</li> </ul> |
| P0280-11 | Cylinder 7 Injector                                                | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       |

|          |                                                                              |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Circuit High -<br>Circuit short to ground                                    | <ul style="list-style-type: none"> <li>■ Fuel injector circuit, short circuit to ground</li> <li>■ Fuel injector failure</li> </ul>                                                                                                                                                  | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check fuel injector resistance</li> <li>■ Check and install new fuel injector as required</li> </ul>                                                                                                                    |
| P0280-12 | Cylinder 7<br>Injector<br>Circuit High -<br>Circuit short to battery         | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Fuel injector circuit, short circuit to power</li> <li>■ Fuel injector failure</li> </ul>                                                        | <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check fuel injector resistance</li> <li>■ Check and install new fuel injector as required</li> </ul>                      |
| P0280-1C | Cylinder 7<br>Injector "A"<br>Circuit High -<br>Circuit voltage out of range | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>■ Check fuel injector resistance</li> <li>■ Check and install a new fuel injector as required</li> </ul> |
| P0282-11 | Cylinder 8<br>Injector                                                       | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul>                                                                                                                                                  | <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                                               |

|          |                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Circuit Low -<br>Circuit short<br>to ground                            | <ul style="list-style-type: none"> <li>▪ Fuel injector circuit, short circuit to ground</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                      |
| P0282-12 | Cylinder 8<br>Injector<br>Circuit Low -<br>Circuit short<br>to battery | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, short circuit to power</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul>        |
| P0282-13 | Cylinder 8<br>Injector "A"<br>Circuit Low -<br>Circuit open            | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> Fault detected by the injector control module's power stage diagnostic then passed to the main central processing unit</p> </div> <ul style="list-style-type: none"> <li>▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector circuit, open circuit, high resistance</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for open circuit, high resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check fuel injector resistance</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P0282-   | Cylinder 8                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|          |                                                                         |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1C       | Injector "A"<br>Circuit Low -<br>Circuit<br>voltage out of<br>range     | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>■ Check fuel injector resistance</li> <li>■ Check and install a new fuel injector as required</li> </ul> |
| P0283-11 | Cylinder 8<br>Injector<br>Circuit High -<br>Circuit short<br>to ground  | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Fuel injector circuit, short circuit to ground</li> <li>■ Fuel injector failure</li> </ul>                                                       | <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check fuel injector resistance</li> <li>■ Check and install new fuel injector as required</li> </ul>                     |
| P0283-12 | Cylinder 8<br>Injector<br>Circuit High -<br>Circuit short<br>to battery | <ul style="list-style-type: none"> <li>■ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Fuel injector circuit, short circuit to power</li> <li>■ Fuel injector failure</li> </ul>                                                        | <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to power</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check fuel injector resistance</li> <li>■ Check and install new fuel injector as required</li> </ul>                      |

|          |                                                                                            |                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0283-1C | Cylinder 8<br>Injector "A"<br>Circuit High -<br>Circuit<br>voltage out of<br>range         | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Fuel injector failure</li> </ul>                                                     | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>Check fuel injector resistance</li> <li>Check and install a new fuel injector as required</li> </ul> |
| P0299-00 | Turbocharger<br>/Supercharger<br>A Underboost<br>Condition -<br>No sub type<br>information | <ul style="list-style-type: none"> <li>Air intake system, boost air system high pressure boost air leak</li> <li>Air intake system, low pressure intake blocked or restricted</li> <li>Supercharger rotors worn or damaged</li> <li>Air intake system pressure sensors incorrect readings</li> <li>Supercharger failure</li> </ul> | <ul style="list-style-type: none"> <li>Test air intake system, boost air system for air leaks</li> <li>Check air intake system, low pressure intake for blockages or restriction</li> <li>Check condition of supercharger and rotors</li> <li>Check engine control module for related air intake system pressure sensor DTCs and refer to this DTC index</li> <li>Check and install new supercharger as required</li> </ul>                                                                                 |
| P029D-13 | Cylinder 1-<br>Injector<br>Leaking -<br>Circuit open                                       | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector harness failure</li> </ul>                                                                                                                                                           | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul>                                                                                                                                                                                                                                                               |
| P02A1-13 | Cylinder 2 -<br>Injector<br>Leaking -<br>Circuit open                                      | <ul style="list-style-type: none"> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector harness failure</li> </ul>                                                                                                                                                           | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul>                                                                                                                                                                                                                                                               |
| P02A5-   | Cylinder 3 -                                                                               |                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|          |                                                |                                                                                                                                                                              |                                                                                                                                                                                                                                                   |
|----------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 71       | Injector Leaking - Actuator stuck              | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector harness failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul> |
| P02A9-13 | Cylinder 4 - Injector Leaking - Circuit open   | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector harness failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul> |
| P02AD-13 | Cylinder 5 - Injector Leaking - Circuit open   | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector harness failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul> |
| P02B1-71 | Cylinder 6 - Injector Leaking - Actuator stuck | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector harness failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul> |
| P02B5-71 | Cylinder 7 - Injector Leaking - Actuator stuck | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector harness failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul> |
| P02B9-71 | Cylinder 8 - Injector Leaking - Actuator stuck | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector harness failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity</li> </ul> |
| P02EE-   | Cylinder 1                                     |                                                                                                                                                                              |                                                                                                                                                                                                                                                   |

|          |                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01       | Injector<br>Circuit Range<br>/Performance<br>- General<br>Electrical<br>Failure                 | <ul style="list-style-type: none"> <li>▪ Fuel injector control circuits (high and low) short circuit to each other</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check both the fuel injector control circuits (high and low) for short circuit to each other</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel injector as required</li> </ul>                   |
| P02EE-1C | Cylinder 1<br>Injector<br>Circuit Range<br>/Performance<br>- Circuit<br>voltage out of<br>range | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> The fuel injector control within the engine control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set</p> </div> <ul style="list-style-type: none"> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P02EF-01 | Cylinder 2<br>Injector<br>Circuit Range<br>/Performance<br>- General<br>Electrical<br>Failure   | <ul style="list-style-type: none"> <li>▪ Fuel injector control circuits (high and low) short circuit to each other</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check both the fuel injector control circuits (high and low) for short circuit to each other</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic</li> </ul>                                                                                                                                                        |

|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                     |
| P02EF-1C | Cylinder 2 Injector Circuit Range /Performance - Circuit voltage out of range | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The fuel injector control within the engine control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set</p> </div> <ul style="list-style-type: none"> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul>   | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P02F0-00 | Cylinder 3 Injector Circuit Range /Performance - No sub type information      | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The fuel injector control within the engine control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set</p> </div> <ul style="list-style-type: none"> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P02F0-01 | Cylinder 3 Injector Circuit Range /Performance - General Electrical Failure   | <ul style="list-style-type: none"> <li>▪ Fuel injector control circuits (high and low) short circuit to each other</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check both the fuel injector control circuits (high and low) for short circuit to each other</li> <li>▪ Inspect connectors for signs of water ingress, and</li> </ul>                                                                                                                                                                                                                                                                                              |

|          |                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>pins for damage and/or corrosion</p> <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                             |
| P02F0-1C | Cylinder 3 Injector Circuit Range /Performance<br>- Circuit voltage out of range | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The fuel injector control within the powertrain control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set.</p> </div> <ul style="list-style-type: none"> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check and install a new fuel injector as required</li> </ul>                  |
| P02F1-01 | Cylinder 4 Injector Circuit Range /Performance<br>- General Electrical Failure   | <ul style="list-style-type: none"> <li>▪ Fuel injector control circuits (high and low) short circuit to each other</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check both the fuel injector control circuits (high and low) for short circuit to each other</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P02F1-1C | Cylinder 4 Injector Circuit Range                                                | <div>  <b>NOTE:</b> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                       |

|          |                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | /Performance<br>- Circuit voltage out of range                                   | <p><b>Monitor description.</b> The fuel injector control within the engine control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set</p> <ul style="list-style-type: none"> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector failure</li> </ul>                                                                          | <p>short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel injector as required</li> </ul>                                                              |
| P02F2-01 | Cylinder 5 Injector Circuit Range /Performance<br>- General Electrical Failure   | <ul style="list-style-type: none"> <li>Fuel injector control circuits (high and low) short circuit to each other</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check both the fuel injector control circuits (high and low) for short circuit to each other</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel injector as required</li> </ul> |
| P02F2-1C | Cylinder 5 Injector Circuit Range /Performance<br>- Circuit voltage out of range | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> The fuel injector control within the engine control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set</p> <ul style="list-style-type: none"> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                        |

|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                               | <ul style="list-style-type: none"> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| P02F3-00 | Cylinder 6 Injector Circuit Range /Performance - No sub type information      | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The fuel injector control within the engine control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set</p> </div> <ul style="list-style-type: none"> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel injector as required</li> </ul> |
| P02F3-01 | Cylinder 6 Injector Circuit Range /Performance - General Electrical Failure   | <ul style="list-style-type: none"> <li>Fuel injector control circuits (high and low) short circuit to each other</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check both the fuel injector control circuits (high and low) for short circuit to each other</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel injector as required</li> </ul>                   |
| P02F3-1C | Cylinder 6 Injector Circuit Range /Performance - Circuit voltage out of range | <div>  <b>NOTES:</b> </div> <div> <ul style="list-style-type: none"> <li><b>Monitor description.</b> The fuel injector control within the powertrain control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set.</li> </ul> </div>                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                     |

|          |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                | <ul style="list-style-type: none"> <li>▪ This DTC is set when the powertrain control module has detected a voltage outside of the expected range, but not identified it as too high or too low.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>▪ Check and install a new fuel injector as required</li> </ul>                                                                                                                                                                                                                                                                                                          |
| P02F4-00 | Cylinder 7 Injector Circuit Range /Performance<br>- No sub type information    | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The fuel injector control within the engine control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set</p> </div> <ul style="list-style-type: none"> <li>▪ Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel injector as required</li> </ul> |
| P02F4-01 | Cylinder 7 Injector Circuit Range /Performance<br>- General Electrical Failure | <ul style="list-style-type: none"> <li>▪ Fuel injector control circuits (high and low) short circuit to each other</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check both the fuel injector control circuits (high and low) for short circuit to each other</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check fuel injector resistance</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new fuel injector as required</li> </ul>                   |

|          |                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P02F4-1C | Cylinder 7 Injector Circuit Range /Performance<br>- Circuit voltage out of range | <div>  <b>NOTES:</b> </div> <ul style="list-style-type: none"> <li> <b>Monitor description.</b> The fuel injector control within the powertrain control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set. </li> <li> This DTC is set when the powertrain control module has detected a voltage outside of the expected range, but not identified it as too high or too low. </li> </ul> <ul style="list-style-type: none"> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>Check and install a new fuel injector as required</li> </ul>                                    |
| P02F5-00 | Cylinder 8 Injector Circuit Range /Performance<br>- No sub type information      | <div>  <b>NOTE:</b> </div> <p><b>Monitor description.</b> The fuel injector control within the engine control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set</p> <ul style="list-style-type: none"> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel injector as required</li> </ul> |
| P02F5-01 | Cylinder 8 Injector Circuit Range /Performance<br>- General Electrical Failure   | <ul style="list-style-type: none"> <li>Fuel injector control circuits (high and low) short circuit to each other</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check both the fuel injector control circuits (high and low) for short circuit to each other</li> <li>Inspect connectors for signs of water ingress, and</li> </ul>                                                                                                                                                                                                                                                                                        |

|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>pins for damage and/or corrosion</p> <ul style="list-style-type: none"> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new fuel injector as required</li> </ul>                                                                                                                                                                                                  |
| P02F5-1C | Cylinder 8 Injector Circuit Range /Performance - Circuit voltage out of range | <div>  <b>NOTES:</b> </div> <ul style="list-style-type: none"> <li><b>Monitor description.</b> The fuel injector control within the powertrain control module monitors the current that the fuel injector uses to open, if the expected opening current is not seen within a set time the DTC is set.</li> <li>This DTC is set when the powertrain control module has detected a voltage outside of the expected range, but not identified it as too high or too low.</li> </ul> <ul style="list-style-type: none"> <li>Fuel injector circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Fuel injector failure</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check fuel injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check fuel injector resistance</li> <li>Using the Jaguar Land Rover approved diagnostic equipment, clear the DTCs and retest</li> <li>Check and install a new fuel injector as required</li> </ul>           |
| P0300-00 | Random Misfire Detected - No sub type information                             | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Misfire detection         </div> <ul style="list-style-type: none"> <li>Poor fuel supply</li> <li>Poor fuel quality</li> <li>Fuel air ratio excessively too lean or too rich</li> <li>Catalyst/exhaust system blockage</li> <li>Spark plug(s) fouled or failed</li> <li>Low Cylinder compression</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Check engine control module for related fuelling, ignition coil, injector or individual cylinder misfire DTCs and refer to this DTC index. Rectify these first</li> <li>If this DTC is raised with P0301-00, P0302-00, P0303-00, P0304-00, P0305-00, P0306-00, P0307-00, or P0308-00, then the fuel delivery system and air intake system should be checked and rectified first</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored</li> </ul> |

|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                       | <ul style="list-style-type: none"> <li>▪ Reluctor ring</li> <li>▪ Crankshaft position sensor failure</li> <li>▪ Camshaft position sensor failure</li> <li>▪ Injector or ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Injector circuit short circuit to ground, short circuit to power, open circuit</li> <li>▪ Injector(s) failure</li> </ul>   | <p>DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>▪ Check the fuel system for blockages or restrictions, repair as required</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the catalyst/exhaust system for blockage, repair as required</li> <li>▪ Check and install a new spark plug(s) as required</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Inspect reluctor ring for damage</li> <li>▪ Check and install a new crankshaft position sensor as required</li> <li>▪ Check and install a new camshaft position sensor as required</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Refer to the electrical circuit diagrams and check injector(s) circuit for short circuit to ground, short circuit to power, open circuit</li> <li>▪ Identify the misfiring cylinder. Check and install a new injector as required</li> </ul> |
| P0301-00 | Cylinder 1 Misfire Detected - No sub type information | <ul style="list-style-type: none"> <li>▪ Other engine control module related DTCs</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ignition system</li> <li>▪ Spark plug(s)</li> <li>▪ Ignition coil(s)</li> <li>▪ Crankshaft position sensor</li> <li>▪ Air intake system leakage</li> <li>▪ Low pressure fuel system</li> <li>▪ High pressure fuel system</li> <li>▪ Engine mechanical integrity</li> <li>▪ Low cylinder compression</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs and refer to relevant DTC index. Carry out repair actions for any engine system DTCs before continuing</li> <li>▪ <b>Carry out Misfire pinpoint test. Refer to Electronic Engine Controls V8 S/C 5.0 L - Petrol - Diagnosis and Testing - pinpoint test A - Misfire pinpoint test</b></li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check ignition system</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                       | <ul style="list-style-type: none"> <li>▪ Crank case ventilation system blockages or leakage</li> <li>▪ Reluctor ring</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>▪ Spark plug(s) fouled or failed</li> <li>▪ Ignition coil wiring integrity</li> <li>▪ Crankshaft position sensor wiring integrity</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the low pressure fuel system</li> <li>▪ Check the high pressure fuel system</li> <li>▪ Check engine for mechanical integrity</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Check crank case ventilation system for blockages or leakage</li> <li>▪ Inspect reluctor ring for damage</li> </ul>                                                                                                                                                                                                                                             |
| P0302-00 | Cylinder 2 Misfire Detected - No sub type information | <ul style="list-style-type: none"> <li>▪ Other engine control module related DTCs</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ignition system</li> <li>▪ Spark plug(s)</li> <li>▪ Ignition coil(s)</li> <li>▪ Crankshaft position sensor</li> <li>▪ Air intake system leakage</li> <li>▪ Low pressure fuel system</li> <li>▪ High pressure fuel system</li> <li>▪ Engine mechanical integrity</li> <li>▪ Low cylinder compression</li> <li>▪ Crank case ventilation system blockages or leakage</li> <li>▪ Reluctor ring</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs and refer to relevant DTC index. Carry out repair actions for any engine system DTCs before continuing</li> <li>▪ <b>Carry out Misfire pinpoint test. Refer to Electronic Engine Controls V8 S/C 5.0 L - Petrol - Diagnosis and Testing - pinpoint test A - Misfire pinpoint test</b></li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check ignition system</li> <li>▪ Spark plug(s) fouled or failed</li> <li>▪ Ignition coil wiring integrity</li> <li>▪ Crankshaft position sensor wiring integrity</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the low pressure fuel system</li> <li>▪ Check the high pressure fuel system</li> </ul> |

|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>▪ Check engine for mechanical integrity</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Check crank case ventilation system for blockages or leakage</li> <li>▪ Inspect reluctor ring for damage</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| P0303-00 | Cylinder 3 Misfire Detected - No sub type information | <ul style="list-style-type: none"> <li>▪ Other engine control module related DTCs</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ignition system</li> <li>▪ Spark plug(s)</li> <li>▪ Ignition coil(s)</li> <li>▪ Crankshaft position sensor</li> <li>▪ Air intake system leakage</li> <li>▪ Low pressure fuel system</li> <li>▪ High pressure fuel system</li> <li>▪ Engine mechanical integrity</li> <li>▪ Low cylinder compression</li> <li>▪ Crank case ventilation system blockages or leakage</li> <li>▪ Reluctor ring</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs and refer to relevant DTC index. Carry out repair actions for any engine system DTCs before continuing</li> <li>▪ <b>Carry out Misfire pinpoint test. Refer to Electronic Engine Controls V8 S/C 5.0 L - Petrol - Diagnosis and Testing - pinpoint test A - Misfire pinpoint test</b></li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check ignition system</li> <li>▪ Spark plug(s) fouled or failed</li> <li>▪ Ignition coil wiring integrity</li> <li>▪ Crankshaft position sensor wiring integrity</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the low pressure fuel system</li> <li>▪ Check the high pressure fuel system</li> <li>▪ Check engine for mechanical integrity</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Check crank case ventilation system for blockages or leakage</li> <li>▪ Inspect reluctor ring for damage</li> </ul> |
| P0304-00 | Cylinder 4 Misfire                                    | <ul style="list-style-type: none"> <li>▪ Other engine control module related DTCs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Detected - No sub type information                    | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ignition system</li> <li>▪ Spark plug(s)</li> <li>▪ Ignition coil(s)</li> <li>▪ Crankshaft position sensor</li> <li>▪ Air intake system leakage</li> <li>▪ Low pressure fuel system</li> <li>▪ High pressure fuel system</li> <li>▪ Engine mechanical integrity</li> <li>▪ Low cylinder compression</li> <li>▪ Crank case ventilation system blockages or leakage</li> <li>▪ Reluctor ring</li> </ul> | <p>and refer to relevant DTC index. Carry out repair actions for any engine system DTCs before continuing</p> <ul style="list-style-type: none"> <li>▪ <b>Carry out Misfire pinpoint test. Refer to Electronic Engine Controls V8 S/C 5.0 L - Petrol - Diagnosis and Testing - pinpoint test A - Misfire pinpoint test</b></li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check ignition system</li> <li>▪ Spark plug(s) fouled or failed</li> <li>▪ Ignition coil wiring integrity</li> <li>▪ Crankshaft position sensor wiring integrity</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the low pressure fuel system</li> <li>▪ Check the high pressure fuel system</li> <li>▪ Check engine for mechanical integrity</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Check crank case ventilation system for blockages or leakage</li> <li>▪ Inspect reluctor ring for damage</li> </ul> |
| P0305-00 | Cylinder 5 Misfire Detected - No sub type information | <ul style="list-style-type: none"> <li>▪ Other engine control module related DTCs</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ignition system</li> <li>▪ Spark plug(s)</li> <li>▪ Ignition coil(s)</li> <li>▪ Crankshaft position sensor</li> <li>▪ Air intake system leakage</li> <li>▪ Low pressure fuel system</li> <li>▪ High pressure fuel system</li> <li>▪ Engine mechanical integrity</li> </ul>                                                                        | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs and refer to relevant DTC index. Carry out repair actions for any engine system DTCs before continuing</li> <li>▪ <b>Carry out Misfire pinpoint test. Refer to Electronic Engine Controls V8 S/C 5.0 L - Petrol - Diagnosis and Testing - pinpoint test A - Misfire pinpoint test</b></li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                       | <ul style="list-style-type: none"> <li>▪ Low cylinder compression</li> <li>▪ Crank case ventilation system blockages or leakage</li> <li>▪ Reluctor ring</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li>▪ Check ignition system</li> <li>▪ Spark plug(s) fouled or failed</li> <li>▪ Ignition coil wiring integrity</li> <li>▪ Crankshaft position sensor wiring integrity</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the low pressure fuel system</li> <li>▪ Check the high pressure fuel system</li> <li>▪ Check engine for mechanical integrity</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Check crank case ventilation system for blockages or leakage</li> <li>▪ Inspect reluctor ring for damage</li> </ul>                                                                                                                                                                                                            |
| P0306-00 | Cylinder 6 Misfire Detected - No sub type information | <ul style="list-style-type: none"> <li>▪ Other engine control module related DTCs</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ignition system</li> <li>▪ Spark plug(s)</li> <li>▪ Ignition coil(s)</li> <li>▪ Crankshaft position sensor</li> <li>▪ Air intake system leakage</li> <li>▪ Low pressure fuel system</li> <li>▪ High pressure fuel system</li> <li>▪ Engine mechanical integrity</li> <li>▪ Low cylinder compression</li> <li>▪ Crank case ventilation system blockages or leakage</li> <li>▪ Reluctor ring</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs and refer to relevant DTC index. Carry out repair actions for any engine system DTCs before continuing</li> <li>▪ <b>Carry out Misfire pinpoint test. Refer to Electronic Engine Controls V8 S/C 5.0 L - Petrol - Diagnosis and Testing - pinpoint test A - Misfire pinpoint test</b></li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check ignition system</li> <li>▪ Spark plug(s) fouled or failed</li> <li>▪ Ignition coil wiring integrity</li> <li>▪ Crankshaft position sensor wiring integrity</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the low pressure fuel system</li> <li>▪ Check the high pressure fuel system</li> </ul> |

|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>▪ Check engine for mechanical integrity</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Check crank case ventilation system for blockages or leakage</li> <li>▪ Inspect reluctor ring for damage</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| P0307-00 | Cylinder 7 Misfire Detected - No sub type information | <ul style="list-style-type: none"> <li>▪ Other engine control module related DTCs</li> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ignition system</li> <li>▪ Spark plug(s)</li> <li>▪ Ignition coil(s)</li> <li>▪ Crankshaft position sensor</li> <li>▪ Air intake system leakage</li> <li>▪ Low pressure fuel system</li> <li>▪ High pressure fuel system</li> <li>▪ Engine mechanical integrity</li> <li>▪ Low cylinder compression</li> <li>▪ Crank case ventilation system blockages or leakage</li> <li>▪ Reluctor ring</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs and refer to relevant DTC index. Carry out repair actions for any engine system DTCs before continuing</li> <li>▪ <b>Carry out Misfire pinpoint test. Refer to Electronic Engine Controls V8 S/C 5.0 L - Petrol - Diagnosis and Testing - pinpoint test A - Misfire pinpoint test</b></li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check ignition system</li> <li>▪ Spark plug(s) fouled or failed</li> <li>▪ Ignition coil wiring integrity</li> <li>▪ Crankshaft position sensor wiring integrity</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the low pressure fuel system</li> <li>▪ Check the high pressure fuel system</li> <li>▪ Check engine for mechanical integrity</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Check crank case ventilation system for blockages or leakage</li> <li>▪ Inspect reluctor ring for damage</li> </ul> |
| P0308-00 | Cylinder 8 Misfire                                    | <ul style="list-style-type: none"> <li>▪ Other engine control module related DTCs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>▪ Check engine control module for related DTCs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Detected - No sub type information                                                    | <ul style="list-style-type: none"> <li>▪ Connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>▪ Ignition system</li> <li>▪ Spark plug(s)</li> <li>▪ Ignition coil(s)</li> <li>▪ Crankshaft position sensor</li> <li>▪ Air intake system leakage</li> <li>▪ Low pressure fuel system</li> <li>▪ High pressure fuel system</li> <li>▪ Engine mechanical integrity</li> <li>▪ Low cylinder compression</li> <li>▪ Crank case ventilation system blockages or leakage</li> <li>▪ Reluctor ring</li> </ul> | <p>and refer to relevant DTC index. Carry out repair actions for any engine system DTCs before continuing</p> <ul style="list-style-type: none"> <li>▪ <b>Carry out Misfire pinpoint test. Refer to Electronic Engine Controls V8 S/C 5.0 L - Petrol - Diagnosis and Testing - pinpoint test A - Misfire pinpoint test</b></li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check ignition system</li> <li>▪ Spark plug(s) fouled or failed</li> <li>▪ Ignition coil wiring integrity</li> <li>▪ Crankshaft position sensor wiring integrity</li> <li>▪ Check for air leaks within the air intake system, repair as required</li> <li>▪ Check the low pressure fuel system</li> <li>▪ Check the high pressure fuel system</li> <li>▪ Check engine for mechanical integrity</li> <li>▪ Carry out cylinder compression checks as required</li> <li>▪ Check crank case ventilation system for blockages or leakage</li> <li>▪ Inspect reluctor ring for damage</li> </ul> |
| P0316-00 | Engine Misfire Detected On Startup (First 1000 Revolutions) - No sub type information | <ul style="list-style-type: none"> <li>▪ Other misfire related DTCs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>▪ Check powertrain control module for misfire related DTCs and refer to this DTC index</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| P0324-00 | Knock Control System Error - No sub type information                                  | <ul style="list-style-type: none"> <li>▪ Crankshaft position sensor signal is corrupted</li> <li>▪ Engine control module software failure</li> <li>▪ Engine control module hardware failure</li> </ul>                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>▪ Check engine control module for crankshaft position sensor DTCs and refer to this DTC index</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment check and install latest relevant level of software to the engine control module</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|          |                                                                 |                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                 |                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Check and install a new engine control module as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| P0327-00 | Knock Sensor 1 Circuit Low (Bank1) - No sub type information    | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Knock sensor</li> </ul>                                                                                                     | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul> |
| P0327-11 | Knock Sensor 1 Circuit Low (Bank1) - Circuit short to ground    | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to ground</li> <li>Knock sensor</li> </ul>                                                                                                                                                            | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul>                                                        |
| P0327-21 | Knock Sensor 1 Circuit Low (Bank1) - Signal amplitude < minimum | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Plausibility of sensor output against engine speed </div> <ul style="list-style-type: none"> <li>Engine misfire</li> </ul> | <ul style="list-style-type: none"> <li>Check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first</li> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power</li> </ul>                                                                                                                                                                                                                                                                                                                                            |

|          |                                                                 |                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                 | <ul style="list-style-type: none"> <li>▪ Knock sensor circuit short circuit to ground, short circuit to power</li> <li>▪ Knock sensor</li> </ul>                                | <ul style="list-style-type: none"> <li>▪ Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>▪ 1 to 1.8 Mega Ohms</li> </ul> </li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new knock sensor as required</li> </ul>                                                                                                                                                                       |
| P0328-00 | Knock Sensor 1 Circuit High (Bank 1) - No sub type information  | <ul style="list-style-type: none"> <li>▪ Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Knock sensor</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>▪ 1 to 1.8 Mega Ohms</li> </ul> </li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new knock sensor as required</li> </ul> |
| P0328-12 | Knock Sensor 1 Circuit High (Bank 1) - Circuit short to battery | <ul style="list-style-type: none"> <li>▪ Knock sensor circuit short circuit to power</li> <li>▪ Knock sensor</li> </ul>                                                         | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to power</li> <li>▪ Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>▪ 1 to 1.8 Mega Ohms</li> </ul> </li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                   |

|          |                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Check and install new knock sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P0328-22 | Knock Sensor 1 Circuit High (Bank 1) - Signal amplitude > maximum | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/>           Plausibility of sensor output against engine speed         </div> <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to power</li> <li>Knock sensor</li> <li>Excessive mechanical noise from the engine</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to power</li> <li>Measure knock sensor circuit for correct resistance               <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check for abnormal engine noise that may corrupt the knock sensor signal</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul> |
| P032C-00 | Knock Sensor 3 Circuit Low (Bank1) - No sub type information      | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Knock sensor</li> </ul>                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Measure knock sensor circuit for correct resistance               <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul>                           |
| P032C-11 | Knock Sensor 3 Circuit Low (Bank1) - Circuit short to ground      | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to ground</li> <li>Knock sensor</li> </ul>                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground</li> <li>Measure knock sensor circuit for correct resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|          |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>▪ 1 to 1.8 Mega Ohms</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new knock sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                  |
| P032C-21 | Knock Sensor 3 Circuit Low (Bank1) - Signal amplitude < minimum | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Plausibility of sensor output against engine speed </div> <ul style="list-style-type: none"> <li>▪ Engine misfire</li> <li>▪ Knock sensor circuit short circuit to ground, short circuit to power</li> <li>▪ Knock sensor</li> </ul> | <ul style="list-style-type: none"> <li>▪ Check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first</li> <li>▪ Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power</li> <li>▪ Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>▪ 1 to 1.8 Mega Ohms</li> </ul> </li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new knock sensor as required</li> </ul> |
| P032D-00 | Knock Sensor 3 Circuit High (Bank1) - No sub type information   | <ul style="list-style-type: none"> <li>▪ Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Knock sensor</li> </ul>                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>▪ Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>▪ 1 to 1.8 Mega Ohms</li> </ul> </li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Using the Jaguar Land Rover approved diagnostic</li> </ul>                                                                                                                                                                                                                       |

|          |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                        | <p>equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>Check and install new knock sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| P032D-12 | Knock Sensor 3 Circuit High (Bank1) - Circuit short to battery   | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to power</li> <li>Knock sensor</li> </ul>                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to power</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul>                                                                                   |
| P032D-22 | Knock Sensor 3 Circuit High (Bank1) - Signal amplitude > maximum | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Plausibility of sensor output against engine speed </div> <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to power</li> <li>Knock sensor</li> <li>Excessive mechanical noise from the engine</li> </ul> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to power</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check for abnormal engine noise that may corrupt the knock sensor signal</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul> |
| P0332-00 | Knock Sensor 2 Circuit Low (Bank 2) - No sub type information    | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Knock sensor</li> </ul>                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|          |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul>                                                                                                                   |
| P0332-11 | Knock Sensor 2 Circuit Low (Bank 2) - Circuit short to ground    | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to ground</li> <li>Knock sensor</li> </ul>                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul>      |
| P0332-21 | Knock Sensor 2 Circuit Low (Bank 2) - Signal amplitude < minimum | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Plausibility of sensor output against engine speed </div> <ul style="list-style-type: none"> <li>Engine misfire</li> <li>Knock sensor circuit short circuit to ground, short circuit to power</li> <li>Knock sensor</li> </ul> | <ul style="list-style-type: none"> <li>Check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first</li> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic</li> </ul> |

|          |                                                                   |                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                   |                                                                                                                                                                                                                | <p>equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</p> <ul style="list-style-type: none"> <li>Check and install new knock sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| P0333-00 | Knock Sensor 2 Circuit High (Bank 2) - No sub type information    | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Knock sensor</li> </ul>                                    | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul> |
| P0333-12 | Knock Sensor 2 Circuit High (Bank 2) - Circuit short to battery   | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to power</li> <li>Knock sensor</li> </ul>                                                                                            | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to power</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul>                                                         |
| P0333-22 | Knock Sensor 2 Circuit High (Bank 2) - Signal amplitude > maximum | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Plausibility of sensor output against engine speed </div> | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to power</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                   | <ul style="list-style-type: none"> <li>▪ Knock sensor circuit short circuit to power</li> <li>▪ Knock sensor</li> <li>▪ Excessive mechanical noise from the engine</li> </ul>                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check for abnormal engine noise that may corrupt the knock sensor signal</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>▪ Check and install new knock sensor as required</li> </ul>                                                           |
| P0335-00 | Crankshaft Position Sensor A Circuit - No sub type information                    | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Missing frequency input signal detected from the crankshaft position sensor</p> </div> <ul style="list-style-type: none"> <li>▪ Crankshaft position sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Crankshaft position sensor air gap to target rotor excessive</li> <li>▪ Crankshaft position sensor failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Refer to electrical circuit diagrams and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check crankshaft position sensor air gap to target rotor is correct</li> <li>▪ Check and install a new crankshaft position sensor as required</li> </ul> |
| P0336-00 | Crankshaft Position Sensor A Circuit Range /Performance - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Disturbance detected on the crankshaft frequency input signal</p> </div> <ul style="list-style-type: none"> <li>▪ Crankshaft position sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Crankshaft position sensor air gap to target rotor excessive</li> <li>▪ Crankshaft position sensor failure</li> </ul>             | <ul style="list-style-type: none"> <li>▪ Refer to electrical circuit diagrams and check the crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>▪ Check crankshaft position sensor air gap to target rotor is correct</li> <li>▪ Check and install a new crankshaft position sensor as required</li> </ul> |
| P0336-03 | Crankshaft Position Sensor A Circuit Range /Performance - FM                      | <ul style="list-style-type: none"> <li>▪ Crankshaft position sensor circuit, short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>▪ Refer to electrical circuit diagrams and check the crankshaft position sensor circuit for short circuit to</li> </ul>                                                                                                                                                                                                                                                                                                                       |

|          |                                                               |                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | (Frequency Modulated) / PWM (Pulse Width Modulated) Failures  | <ul style="list-style-type: none"> <li>■ Crankshaft position sensor air gap to target rotor excessive</li> <li>■ Crankshaft position sensor failure</li> </ul>                  | <p>ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Check crankshaft position sensor air gap to target rotor is correct</li> <li>■ Check and install a new crankshaft position sensor as required</li> </ul>                                                                                                                                                                                                                                                                               |
| P033C-00 | Knock Sensor 4 Circuit Low (Bank 2) - No sub type information | <ul style="list-style-type: none"> <li>■ Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Knock sensor</li> </ul> | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>■ Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>■ 1 to 1.8 Mega Ohms</li> </ul> </li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new knock sensor as required</li> </ul> |
| P033C-11 | Knock Sensor 4 Circuit Low (Bank 2) - Circuit short to ground | <ul style="list-style-type: none"> <li>■ Knock sensor circuit short circuit to ground</li> <li>■ Knock sensor</li> </ul>                                                        | <ul style="list-style-type: none"> <li>■ Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground</li> <li>■ Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>■ 1 to 1.8 Mega Ohms</li> </ul> </li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>■ Check and install new knock sensor as required</li> </ul>                                                        |

|          |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P033C-21 | Knock Sensor 4 Circuit Low (Bank 2) - Signal amplitude < minimum | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b><br/> Plausibility of sensor output against engine speed </div> <ul style="list-style-type: none"> <li>Engine misfire</li> <li>Knock sensor circuit short circuit to ground, short circuit to power</li> <li>Knock sensor</li> </ul> | <ul style="list-style-type: none"> <li>Check engine control module for related misfire DTCs and refer to this DTC index. Rectify these first</li> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul> |
| P033D-00 | Knock Sensor 4 Circuit High (Bank 2) - No sub type information   | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Knock sensor</li> </ul>                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul>                                                                                 |
| P033D-12 | Knock Sensor 4 Circuit High (Bank 2) - Circuit short to battery  | <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to power</li> <li>Knock sensor</li> </ul>                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to power</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul>                                                                                                                                                                                                                                                                                                                                  |
| P033D-22 | Knock Sensor 4 Circuit High (Bank 2) - Signal amplitude > maximum                                         | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b><br/>Plausibility of sensor output against engine speed</p> </div> <ul style="list-style-type: none"> <li>Knock sensor circuit short circuit to power</li> <li>Knock sensor</li> <li>Excessive mechanical noise from the engine</li> </ul>                                                                                                      | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check knock sensor circuit for short circuit to power</li> <li>Measure knock sensor circuit for correct resistance <ul style="list-style-type: none"> <li>1 to 1.8 Mega Ohms</li> </ul> </li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check for abnormal engine noise that may corrupt the knock sensor signal</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> <li>Check and install new knock sensor as required</li> </ul> |
| P0341-00 | Camshaft Position Sensor A Circuit Range /Performance (Bank 1 or single sensor) - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b><br/>Disturbance detected on the camshaft input signal</p> </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul> | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul>                                                                                                                                                                                          |
| P0342-00 | Camshaft Position Sensor A Circuit Low (Bank 1 or                                                         | <div>  <b>NOTE:</b> </div>                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|          |                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | single sensor)<br>- No sub type information                                                    | <p><b>Monitor description.</b> Missing input signal detected from the camshaft position sensor. No rise or falling edge detected from the camshaft position sensor. Input signal from the camshaft position sensor is low</p> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul>                                                                                                        | <p>camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul>                                                                                                                     |
| P0343-00 | Camshaft Position Sensor A Circuit High (Bank 1 or single sensor)<br>- No sub type information | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Missing input signal detected from the camshaft position sensor. No rise or falling edge detected from the camshaft position sensor. Input signal from the camshaft position sensor is high</p> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul> | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul> |
| P0346-00 | Camshaft Position Sensor A Circuit Range /Performance (Bank 2) - No sub type information       | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Disturbance detected on the camshaft input signal</p> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul>                                                                                                                                         | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul> |

|          |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0347-00 | Camshaft Position Sensor A Circuit Low (Bank 2) - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Missing input signal detected from the camshaft position sensor. No rise or falling edge detected from the camshaft position sensor. Input signal from the camshaft position sensor is low         </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul>   | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul> |
| P0348-00 | Camshaft Position Sensor A Circuit High (Bank 2) - No sub type information | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> Missing input signal detected from the camshaft position sensor. No rise or falling edge detected from the camshaft position sensor. Input signal from the camshaft position sensor is high         </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul> | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul> |
| P0350-04 | Ignition Coil Primary /Secondary Circuit - System Internal Failures        | <div>  <b>NOTE:</b> </div> <div> <b>Monitor description.</b> The main CPU will raise the DTC if no messages are received from the ignition control module (internal to ECM)         </div> <ul style="list-style-type: none"> <li>Power and ground connections to the engine control module open circuit</li> <li>Engine control module failure</li> </ul>                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Vehicle Conditions to enable DTC Logging strategy             <ul style="list-style-type: none"> <li>Ignition on</li> </ul> </li> <li>Prioritised Checks to Perform</li> <li>Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>Refer to the electrical circuit diagrams and check the power and ground connections to the module</li> </ul>                           |

|          |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>▪ Check and install new engine control module as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| P0350-64 | Ignition Coil Primary /Secondary Circuit - Signal plausibility failure  | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The main CPU will raise the DTC if no messages are received from the ignition control module (internal to ECM)</p> </div> <ul style="list-style-type: none"> <li>▪ Power and ground connections to the engine control module open circuit</li> <li>▪ Engine control module failure</li> </ul> | <ul style="list-style-type: none"> <li>▪ Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>▪ Ignition on</li> </ul> </li> <li>▪ Prioritised Checks to Perform</li> <li>▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> <li>▪ Refer to the electrical circuit diagrams and check the power and ground connections to the module</li> <li>▪ Check and install new engine control module as required</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> <li>▪ Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0350-85 | Ignition Coil Primary /Secondary Circuit - Signal above allowable range | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> The main CPU will raise the DTC if no messages are received from the ignition control module (internal to ECM)</p> </div> <ul style="list-style-type: none"> <li>▪ Power and ground connections to the engine control module open circuit</li> </ul>                                        | <ul style="list-style-type: none"> <li>▪ Vehicle Conditions to enable DTC Logging strategy <ul style="list-style-type: none"> <li>▪ Ignition on</li> </ul> </li> <li>▪ Prioritised Checks to Perform</li> <li>▪ Check engine control module for additional DTCs and refer to relevant DTC index. Rectify these first</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                 | <ul style="list-style-type: none"> <li>Engine control module failure</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check the power and ground connections to the module</li> <li>Check and install new engine control module as required</li> <li>Using the Jaguar Land Rover approved diagnostic equipment carry out routine "Configure new modules - Powertrain control module"</li> <li>Using the Jaguar Land Rover approved diagnostic equipment clear all stored DTCs using the 'Diagnosis Menu' tab and retest</li> </ul> |
| P0351-13 | Ignition Coil "A" Primary Control Circuit / Open - Circuit open | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> Open circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC</p> </div> <ul style="list-style-type: none"> <li>Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Ignition coil fuse supply failure</li> <li>Ignition coil circuit open circuit</li> </ul>   | <ul style="list-style-type: none"> <li>Check ignition coil connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check ignition coil circuit for open circuit</li> </ul>                                                                                                                                                             |
| P0352-13 | Ignition Coil "B" Primary Control Circuit / Open - Circuit open | <div>  <b>NOTE:</b> <p><b>Monitor description.</b> Open circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC</p> </div> <ul style="list-style-type: none"> <li>Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Ignition coil fuse supply failure</li> <li>Ignition coil circuit open circuit</li> </ul> | <ul style="list-style-type: none"> <li>Check ignition coil connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check ignition coil circuit for open circuit</li> </ul>                                                                                                                                                             |
| P0353-13 | Ignition Coil "C" Primary Control Circuit                       | <div>  <b>NOTE:</b> </div>                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|          |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                            |
|----------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | / Open -<br>Circuit open                                        | <p><b>Monitor description.</b> Open circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC</p> <ul style="list-style-type: none"> <li>■ Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Ignition coil fuse supply failure</li> <li>■ Ignition coil circuit open circuit</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li>■ Check ignition coil connector is not disconnected, connector pin is not backed out</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check ignition coil circuit for open circuit</li> </ul> |
| P0354-13 | Ignition Coil "D" Primary Control Circuit / Open - Circuit open | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Open circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC</p> <ul style="list-style-type: none"> <li>■ Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Ignition coil fuse supply failure</li> <li>■ Ignition coil circuit open circuit</li> </ul>   | <ul style="list-style-type: none"> <li>■ Check ignition coil connector is not disconnected, connector pin is not backed out</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check ignition coil circuit for open circuit</li> </ul> |
| P0355-13 | Ignition Coil "E" Primary Control Circuit / Open - Circuit open | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Open circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC</p> <ul style="list-style-type: none"> <li>■ Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>■ Ignition coil fuse supply failure</li> <li>■ Ignition coil circuit open circuit</li> </ul> | <ul style="list-style-type: none"> <li>■ Check ignition coil connector is not disconnected, connector pin is not backed out</li> <li>■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>■ Refer to the electrical circuit diagrams and check ignition coil circuit for open circuit</li> </ul> |
| P0356-13 | Ignition Coil "F" Primary Control Circuit / Open - Circuit open | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Open circuit detected by the ignition control module, passed to</p>                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li>■ Check ignition coil connector is not disconnected, connector pin is not backed out</li> <li>■ Inspect connectors for signs of water ingress, and</li> </ul>                                                                                                                                       |

|          |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                                          | <p>main CPU within engine control module and sets the DTC</p> <ul style="list-style-type: none"> <li>Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Ignition coil fuse supply failure</li> <li>Ignition coil circuit open circuit</li> </ul>                                                                                                                                                                                                     | <p>pins for damage and/or corrosion</p> <ul style="list-style-type: none"> <li>Refer to the electrical circuit diagrams and check ignition coil circuit for open circuit</li> </ul>                                                                                                                                                  |
| P0357-13 | Ignition Coil "G" Primary Control Circuit / Open - Circuit open                          | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Open circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC</p> <ul style="list-style-type: none"> <li>Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Ignition coil fuse supply failure</li> <li>Ignition coil circuit open circuit</li> </ul>   | <ul style="list-style-type: none"> <li>Check ignition coil connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check ignition coil circuit for open circuit</li> </ul> |
| P0358-13 | Ignition Coil "H" Primary Control Circuit / Open - Circuit open                          | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Open circuit detected by the ignition control module, passed to main CPU within engine control module and sets the DTC</p> <ul style="list-style-type: none"> <li>Ignition coil connector is disconnected, connector pin is backed out, connector pin corrosion</li> <li>Ignition coil fuse supply failure</li> <li>Ignition coil circuit open circuit</li> </ul> | <ul style="list-style-type: none"> <li>Check ignition coil connector is not disconnected, connector pin is not backed out</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Refer to the electrical circuit diagrams and check ignition coil circuit for open circuit</li> </ul> |
| P0366-00 | Camshaft Position Sensor B Circuit Range /Performance (Bank 1) - No sub type information | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Disturbance detected on the camshaft input signal</p> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> </ul>                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> </ul>                                              |

|          |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                                                            | <ul style="list-style-type: none"> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul>                                                                                                                                                                                                                                                |
| P0367-00 | Camshaft Position Sensor B Circuit Low (Bank 1) - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Missing input signal detected from the camshaft position sensor. No rise or falling edge detected from the camshaft position sensor. Input signal from the camshaft position sensor is low</p> </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul>    | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul> |
| P0368-00 | Camshaft Position Sensor B Circuit High (Bank 1) - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Missing input signal detected from the camshaft position sensor. No rise or falling edge detected from the camshaft position sensor. Input signal from the camshaft position sensor is high</p> </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul> | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul> |
| P0391-00 | Camshaft Position Sensor B Circuit Range /Performance                      | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Disturbance detected on the camshaft input signal</p> </div>                                                                                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to</li> </ul>                                                                                                                                                                                                                                                |

|          |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | (Bank 2) - No sub type information                                         | <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul>                                                                                                                                                                                                                                                                                                                                                           | <p>ground, short circuit to power, open circuit, high resistance</p> <ul style="list-style-type: none"> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul>                                                                                                                                                                           |
| P0392-00 | Camshaft Position Sensor B Circuit Low (Bank 2) - No sub type information  | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Missing input signal detected from the camshaft position sensor. No rise or falling edge detected from the camshaft position sensor. Input signal from the camshaft position sensor is low</p> </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul>    | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul> |
| P0393-00 | Camshaft Position Sensor B Circuit High (Bank 2) - No sub type information | <div>  <b>NOTE:</b> </div> <div> <p><b>Monitor description.</b> Missing input signal detected from the camshaft position sensor. No rise or falling edge detected from the camshaft position sensor. Input signal from the camshaft position sensor is high</p> </div> <ul style="list-style-type: none"> <li>Loose camshaft position sensor</li> <li>Camshaft position sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Camshaft position sensor reluctor ring air gap excessive</li> </ul> | <ul style="list-style-type: none"> <li>Check camshaft position sensor is installed correctly</li> <li>Refer to the electrical circuit diagrams and check camshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance</li> <li>Inspect connectors for signs of water ingress, and pins for damage and/or corrosion</li> <li>Check reluctor ring to sensor run-out and air gap are within specification</li> </ul> |
| P0420-00 | Catalyst System Efficiency                                                 | <div>  <b>NOTE:</b> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment, check engine</li> </ul>                                                                                                                                                                                                                                                                                                                                                |

|          |                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Below Threshold (Bank 1) - No sub type information                            | <p><b>Monitor description.</b> Oxygen storage capacity of the catalyst too low</p> <ul style="list-style-type: none"> <li>Exhaust system leakage</li> <li>Catalytic converter failed, caused by lean combustion</li> <li>Catalytic converter failed, caused by misfire</li> <li>Catalytic converter failed, caused by excessive oil consumption</li> <li>Catalytic converter failed, caused by contaminated fuel</li> </ul>                                                                                                       | <p>control module, for related DTCs and refer to the relevant DTC index</p> <ul style="list-style-type: none"> <li>Check exhaust system for leakage</li> <li>Refer to the relevant sections of the workshop manual and check the induction system for air leaks</li> <li>Refer to the relevant sections of the workshop manual and check the ignition system for causes of misfire</li> <li>Refer to the relevant sections of the workshop manual and carry out oil consumption checks</li> <li>Confirm with the customer that the correct fuel has been used</li> </ul>                                                                           |
| P0430-00 | Catalyst System Efficiency Below Threshold (Bank 2) - No sub type information | <p> <b>NOTE:</b></p> <p><b>Monitor description.</b> Oxygen storage capacity of the catalyst too low</p> <ul style="list-style-type: none"> <li>Exhaust system leakage</li> <li>Catalytic converter failed, caused by lean combustion</li> <li>Catalytic converter failed, caused by misfire</li> <li>Catalytic converter failed, caused by excessive oil consumption</li> <li>Catalytic converter failed, caused by contaminated fuel</li> </ul> | <ul style="list-style-type: none"> <li>Using the Jaguar Land Rover approved diagnostic equipment, check engine control module, for related DTCs and refer to the relevant DTC index</li> <li>Check exhaust system for leakage</li> <li>Refer to the relevant sections of the workshop manual and check the induction system for air leaks</li> <li>Refer to the relevant sections of the workshop manual and check the ignition system for causes of misfire</li> <li>Refer to the relevant sections of the workshop manual and carry out oil consumption checks</li> <li>Confirm with the customer that the correct fuel has been used</li> </ul> |